

Unit 7 & 8: Part 1

THE FIRM AND ITS CUSTOMERS



Unit 7 & 8

- In Unit 6 we looked at choices that firms make in relation to their workers.
- In the next 4-5 weeks we will look at decisions firms make in relation to their customers (product markets).

Part 1: The firm and its customers / profit maximization

Part 2: Competitive markets

Part 3: Efficiency and fairness

Part 4: Price elasticity of Demand

Part 5: Taxes and price controls

Part 6: Market power and competition policy

Unit 7 & 8: Part 1 - Overview

- Markets
- Product differentiation
- Demand and profit
- Total, average and marginal cost
- Marginal revenue and profit maximization
- Economies and diseconomies of scale
- Monopolies

Key questions

How do firms with **market power** (those in imperfectly competitive markets) interact with their customers?

How do they **choose price** and **quantity** to **maximize their profits**, taking into account the product **demand curve** and the **cost function**?

How do they choose price and quantity? (7.6)

What is the **law of demand** and why does it hold?

How do costs vary with the level of production? (7.3)

What do **marginal revenue** (MR) and **marginal cost** (MC) tell us about profit maximization? (7.6)

What long-run cost advantages may be associated with large-scale production? (7.3)

What are **monopolies** and how do they maximise their profit?

Introduction

Interesting facts



In the 70s

Few thousand workers



3,500 workers



400 workers & 8 stores

40 Years later

108 000 & 300 000 workers

2.3 million workers

147 000 workers & 2 934 stores

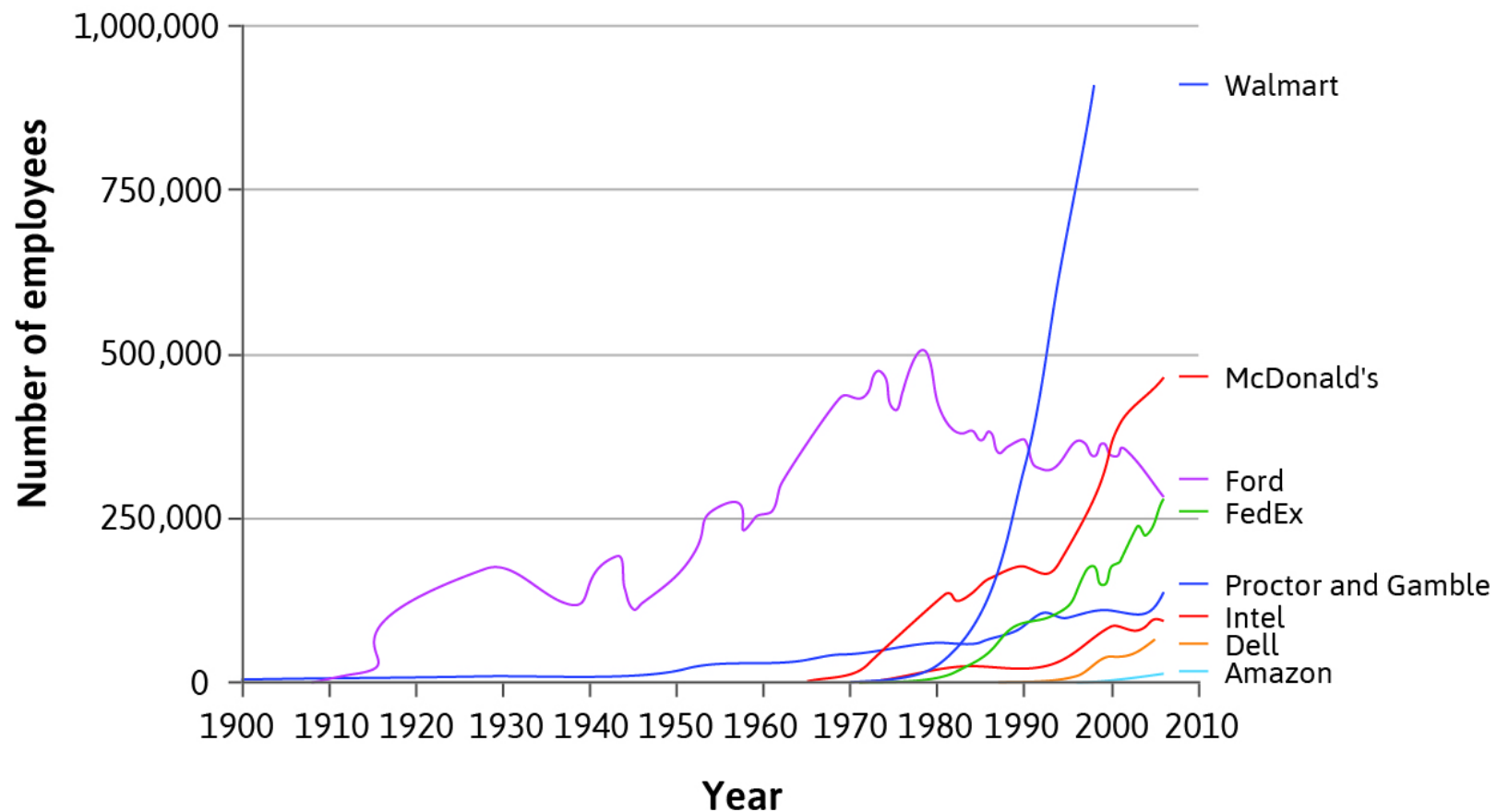


Figure 7.1 Firm size in the US: Number of employees (1900–2006)

In the US, 52% of private-sector employees work in firms with 500+ employees; employees in large firms are also paid more.

This proportion is 40% in SA according to a 2013 study (Kerr et al.); in 2015, those employed in firms with 1000+ workers earned 60% more (on average) than workers in firms of size 100-1000.

How many firms are there in SA?

- There are 267 959 firms in the IRP5 (income tax for employees) data in 2016.
- The Bureau of Economic Research 2016 study estimates the number of small businesses to be in the region of 2.2million (including micro and unregistered businesses).

What strategies can firms use to prosper and grow like the ones in Figure 7.1

- 'Pile it high and sell it cheap,' British Tesco Jack Cohen's motto.
- Apple charge higher prices for iPhones and iPads

What is the difference between Apple and Tesco's strategies ?

But!

Is price the only thing that matters?

What strategies did Shoprite use to achieve growth and prosperity?

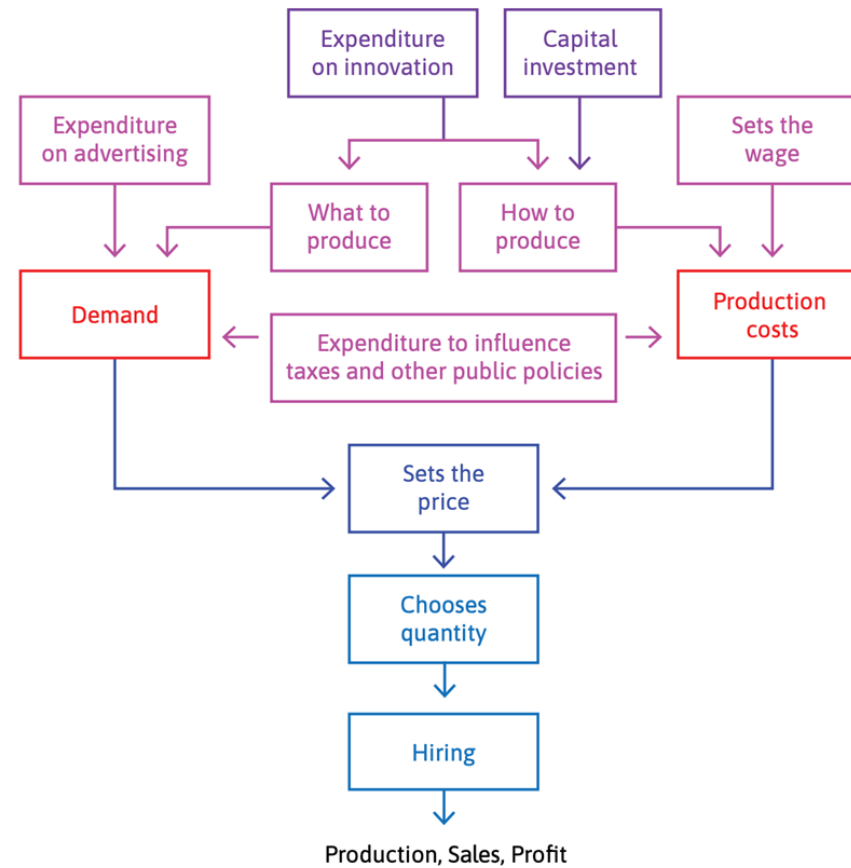


Firms make many different kinds of decisions:

- Financial decisions – e.g. taking out loans, raising financial capital
- Management of assets
- Hiring and wage decisions
- Product innovation, design and choice of product(s)
- Choices about production technologies
- Marketing the product: advertising, branding, pricing, where and how to sell
- Decisions relating to competitors and government

In Unit 7 & 8 we focus on **price** and **quantity** decisions

Fig 7.1 The firm's decision



7.2 Choosing the price

- Unit 7 studies **Price-setting** firms
 - Some firms sell differentiated goods
 - These firms have some market power
 - In some markets there are many firms that sell similar products (but not perfect substitutes)
 - In some markets there is only one seller, a monopoly
 - Such firms have a lot of market power
- Unit 8 studies **Price-taking** firms
 - These firms sell homogeneous (identical) goods
 - These firms have no market power

Markets

- Markets bring buyers and sellers together so that they can transact
- Most markets are not specific physical spaces
- In Economics, a market relates to **one** good or service, e.g. the market for crude oil



Product differentiation

Homogeneous (identical) goods:

- Goods sold by different sellers in the market are completely identical in every respect, including how buyers perceive them
- Impossible to sell at a price higher than other sellers
- Firms are price takers

Differentiated goods:

- Products differ by characteristics, such as quality, style, location, etc.
- Consumers value unique characteristics and variety (e.g. shoes)
- Different firms compete for buyers on multiple fronts – price and quality
- Each firm sets own price
- A higher price means **some** but not all customers will move to competitors

These provide two different **models** for analyzing firms and markets – we choose the model most suited to the question we're trying to answer

7.2 Choosing the price

The firm's objective: profit

$$\text{Profit} = \text{Revenue} - \text{Costs}$$

$$\pi = \text{TR} - \text{TC}$$

Price and Quantity
Demand Curve

Quantity
Cost Curves

7.2 Choosing price: profit

The equations linking revenue, costs and profit are given by:

$$\begin{aligned}\text{Total costs} &= \text{unit cost} \times \text{quantity} \\ \text{TC} &= c \times Q\end{aligned}$$

For now, we will assume that c is constant: every unit produced costs c

$$\begin{aligned}\text{Total revenue} &= \text{price} \times \text{quantity} \\ \text{TR} &= P \times Q\end{aligned}$$

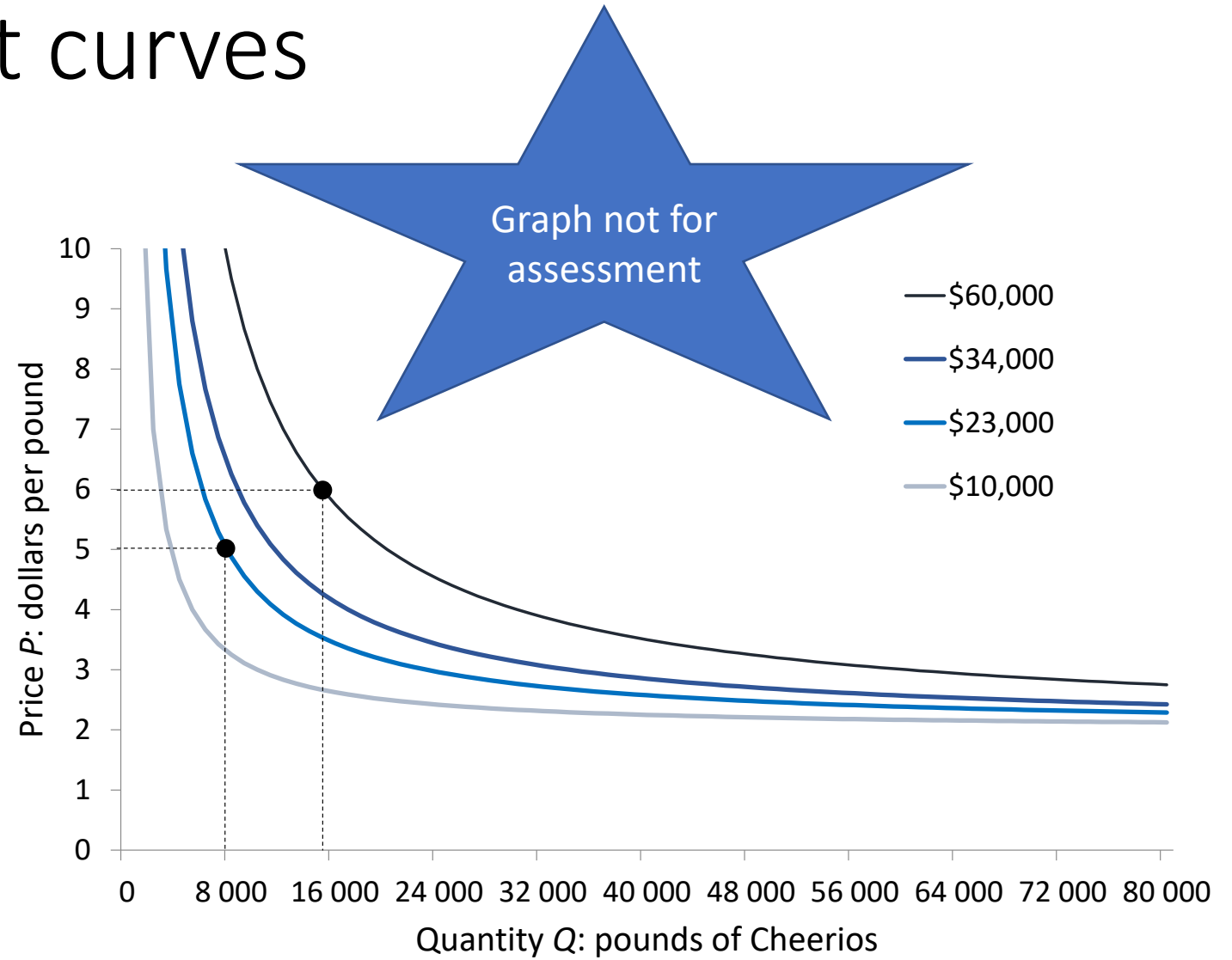
$$\begin{aligned}\text{Profit} &= \text{TR} - \text{TC} \\ \pi &= (P \times Q) - (c \times Q) \\ &= (P - c) \times Q\end{aligned}$$

We can see profit increases if
1) Cost per unit decreases, 2)
Price increases or 3) Quantity
increases (as long as $P > c$)

So, a firm would like to choose high P and Q . Can it choose whatever it wants to?

7.2 A note: isoprofit curves

- The CORE text uses isoprofit curves, which are the firm's indifference curves
- Higher indifference curves = higher profit
- E.g. higher P and higher Q will make the firm better off, thus on a higher indifference curve
- A firm can also compensate for lower P with higher Q and vice versa
- **Isoprofit curves are not included in Economics 114**



7.2 Choosing a price: demand for a differentiated product

In 1996, Jerry Hausman, an economist, used data on weekly sales of family breakfast cereals in US cities to estimate how the weekly quantity of cereal that customers in a typical city would wish to buy would vary with its price per pound (there are 2.2 pounds in 1 kg)

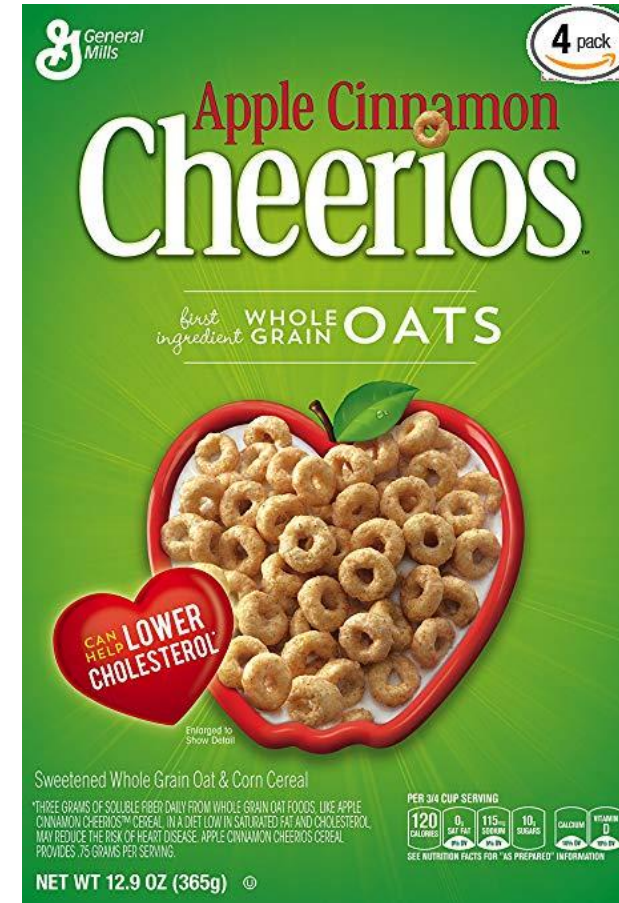
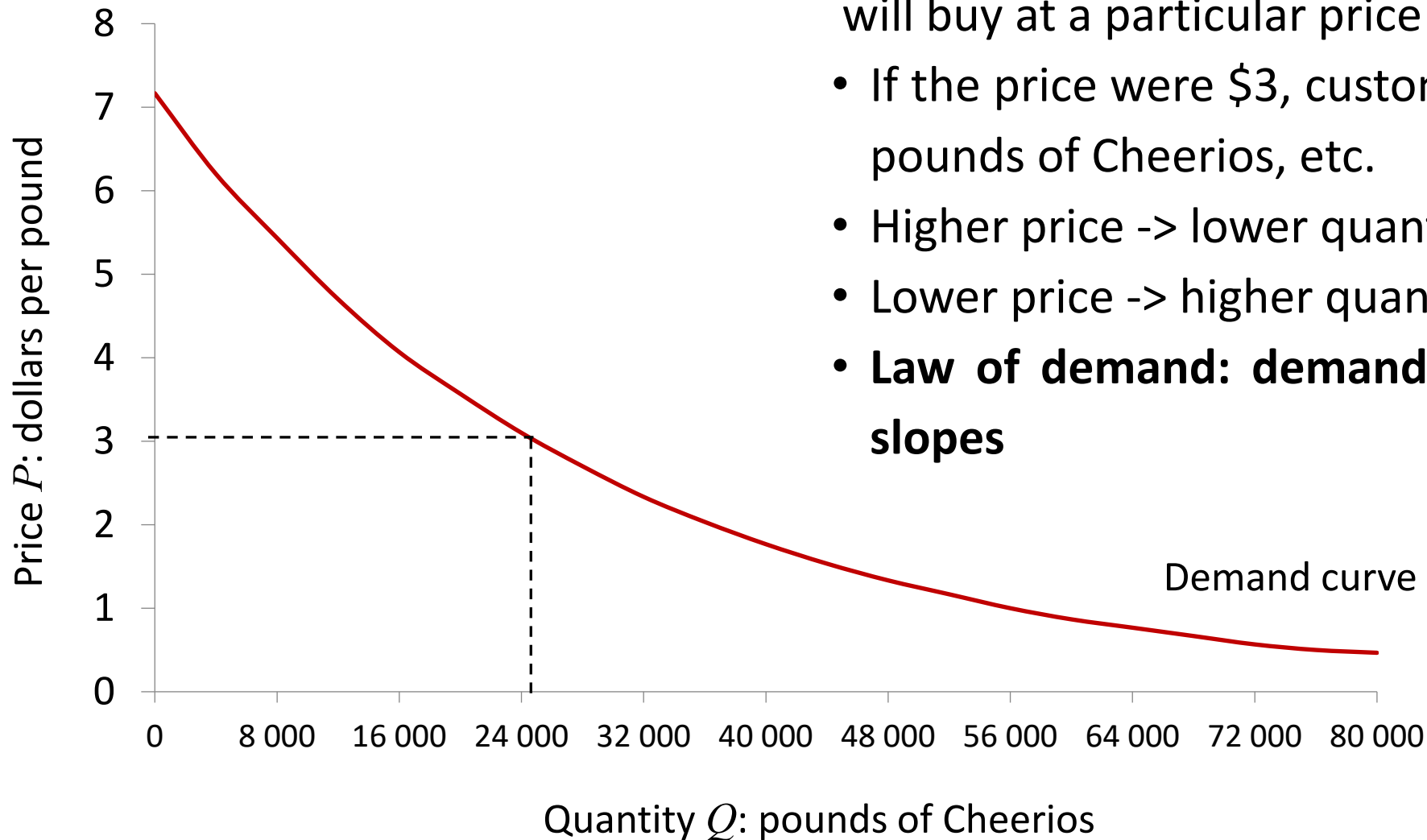


Fig 7.3 Choosing a price: demand



The demand curve shows how much consumers will buy at a particular price

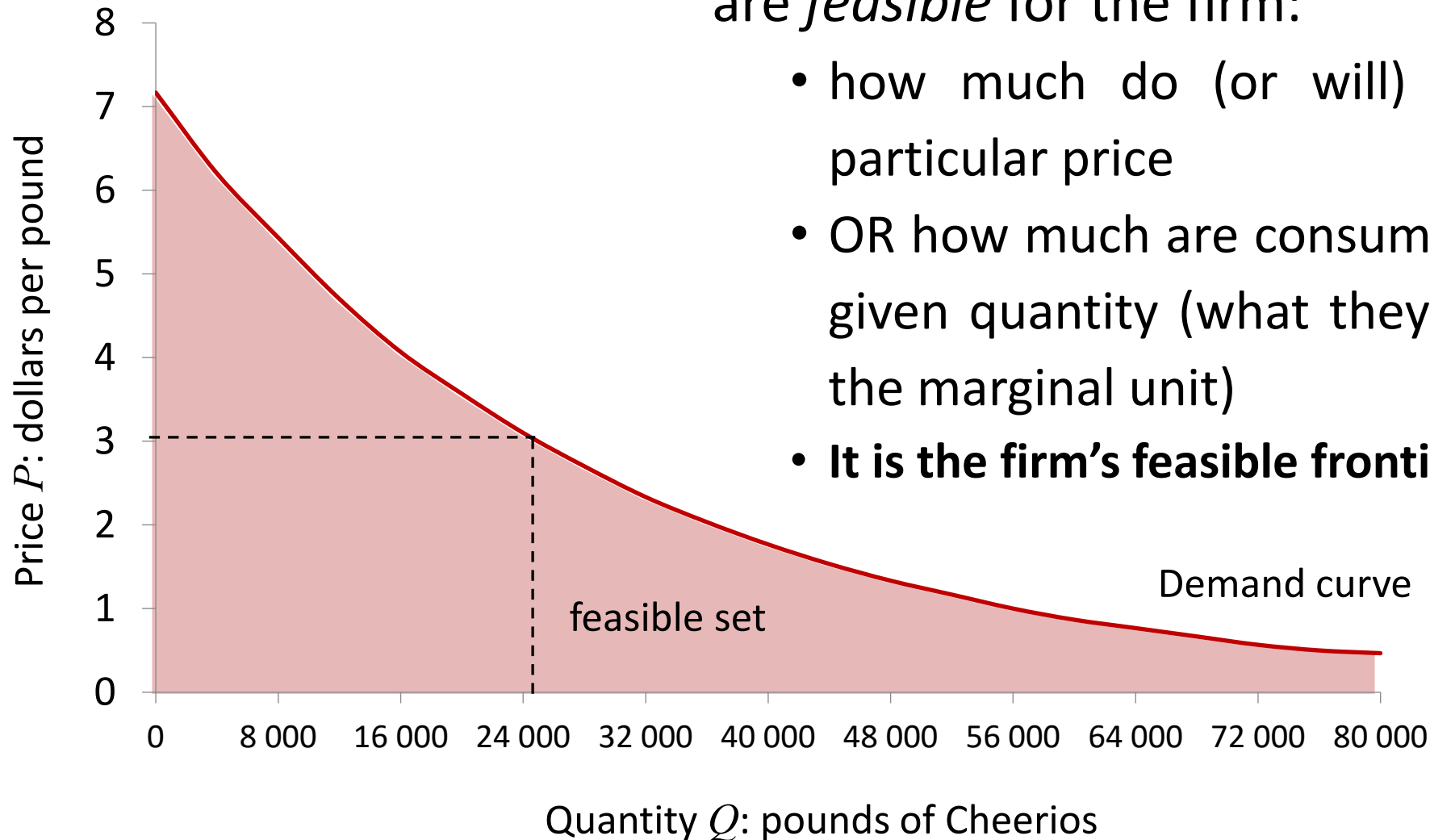
- If the price were \$3, customers demanded 25 000 pounds of Cheerios, etc.
- Higher price $\rightarrow$ lower quantity demanded
- Lower price $\rightarrow$ higher quantity demanded
- **Law of demand: demand curves have negative slopes**

Important: This is the individual **firm's** demand curve

7.2 Choosing a price: demand

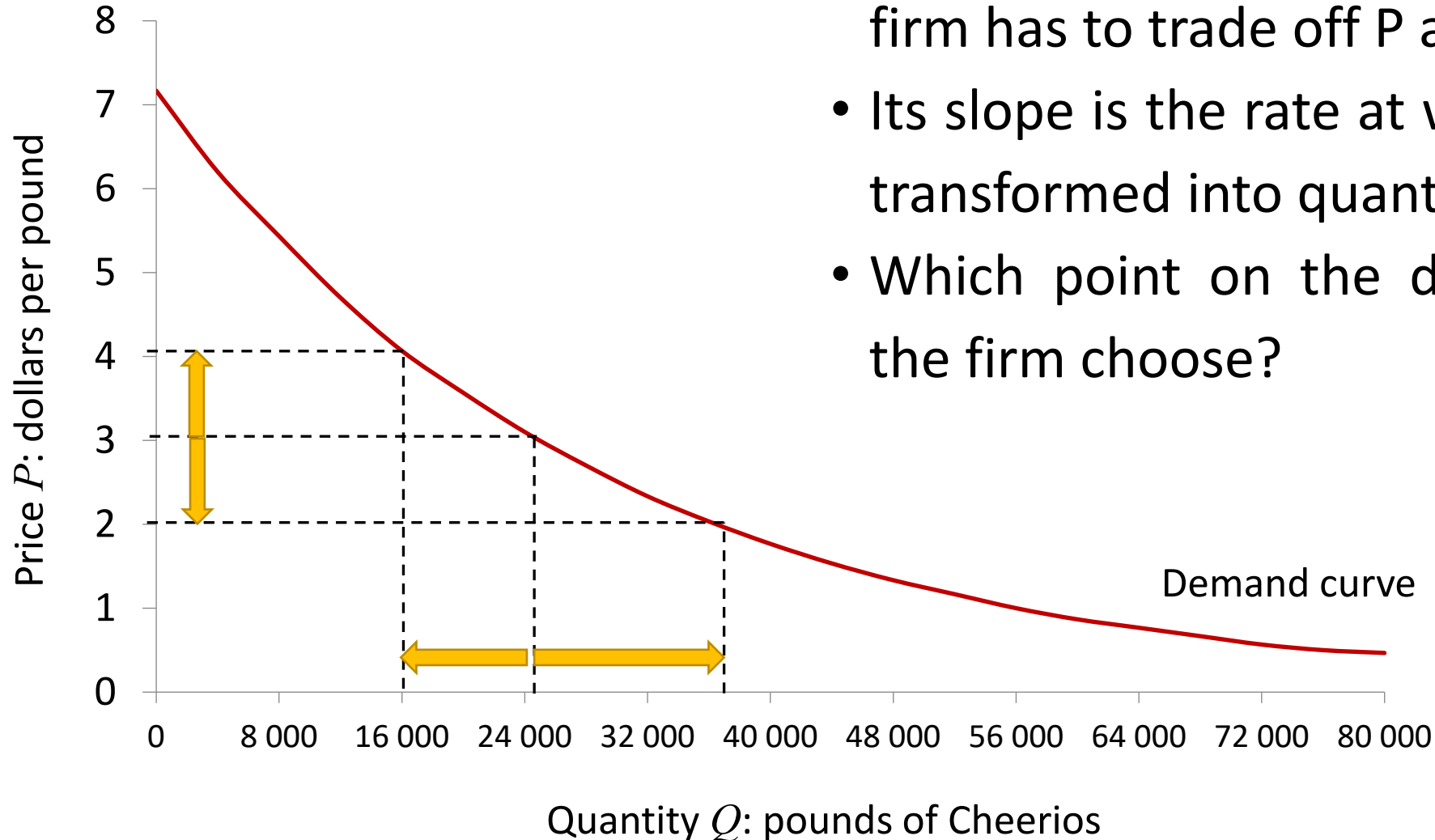
The demand curve shows which P and Q decisions are *feasible* for the firm:

- how much do (or will) consumers buy at a particular price
- OR how much are consumers willing to pay for a given quantity (what they are willing to pay for the marginal unit)
- **It is the firm's feasible frontier (FF)**



7.2 Choosing a price: demand

- The demand curve represents how the firm has to trade off P and Q
- Its slope is the rate at which price can be transformed into quantity
- Which point on the demand curve will the firm choose?

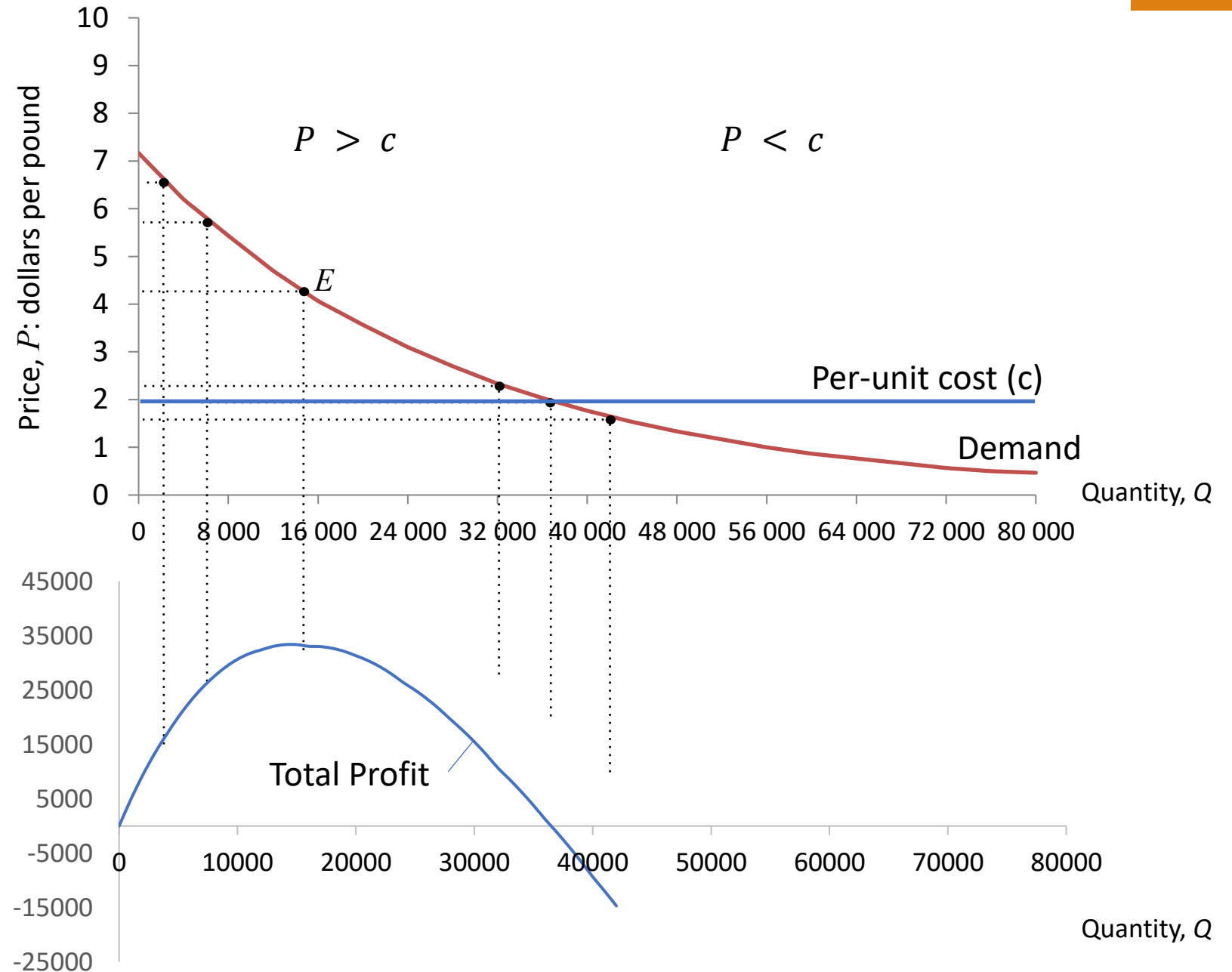


7.2 Choosing a price: profit maximizing choice

- The firm wants to choose a feasible price and quantity combination that will maximise its profit
- The firm could calculate its profit for every price and quantity combination on the demand curve:
 - $\pi = (P - c) \times Q$
 - **For now, we assume constant per-unit cost $c = 2$**
- The profit at each level of P and Q is represented on the profit function

Figure 7.5b. The profit-maximising choice of price and quantity for Apple-Cinnamon Cheerios.

- When Q is low, π is low
- As Q increases, π increases up to a point
- At E , π is maximised
- Beyond E the π decreases
- π decreases to 0 when $P = \$2$
- Beyond \$2, π is negative
 - Why?



Do firms really choose prices this way?

Demand

If you demand something, then you:

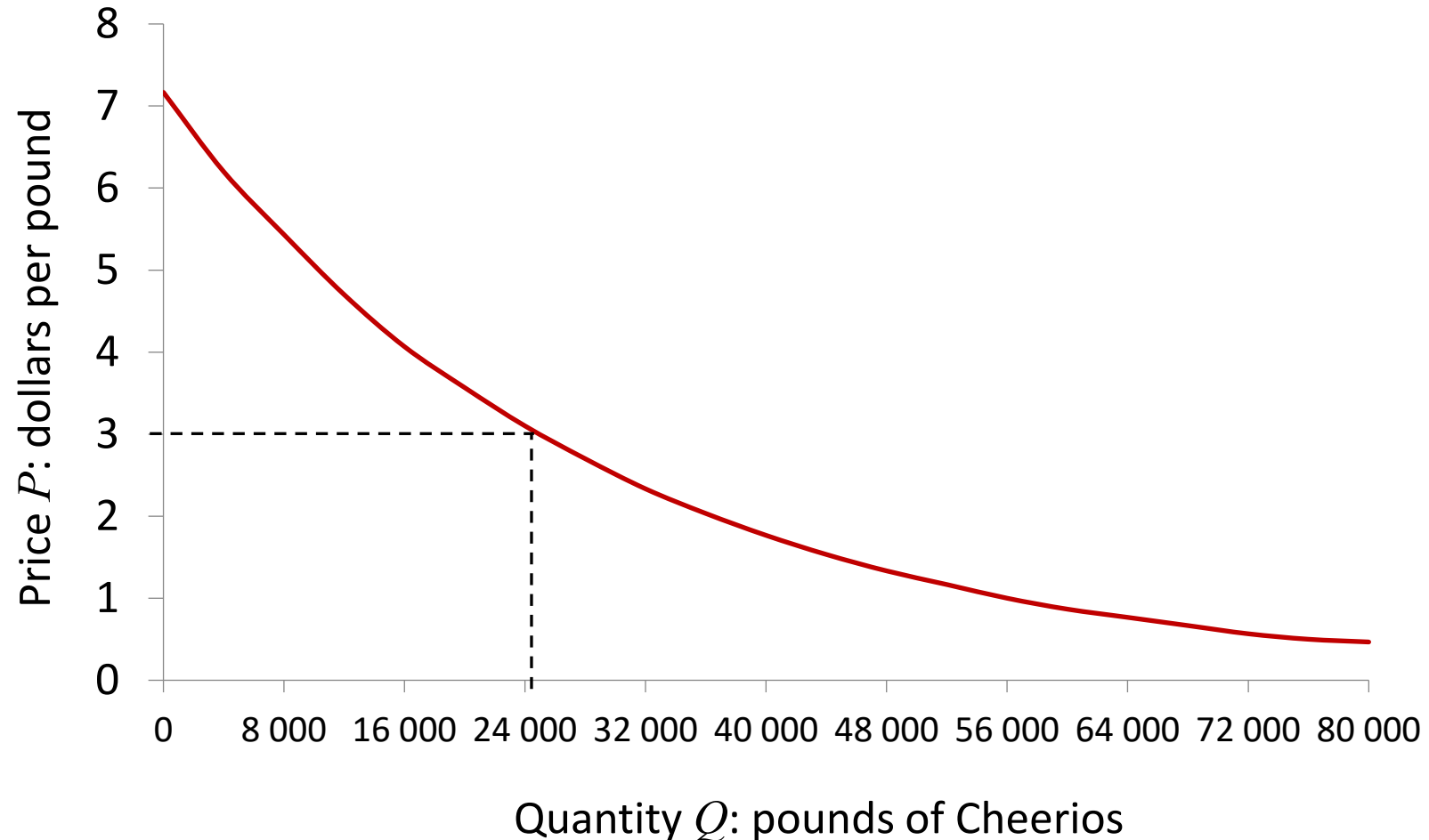
1. Want it
2. Can afford it
3. Plan (decide) to buy it

We now take a closer look at demand:

1. Why does the law of demand hold?
2. Willingness to pay (WTP)
3. Demand from multiple consumers
4. Changes in demand

The law of demand:

There is an **inverse** relationship between the price and the quantity demanded of a product.



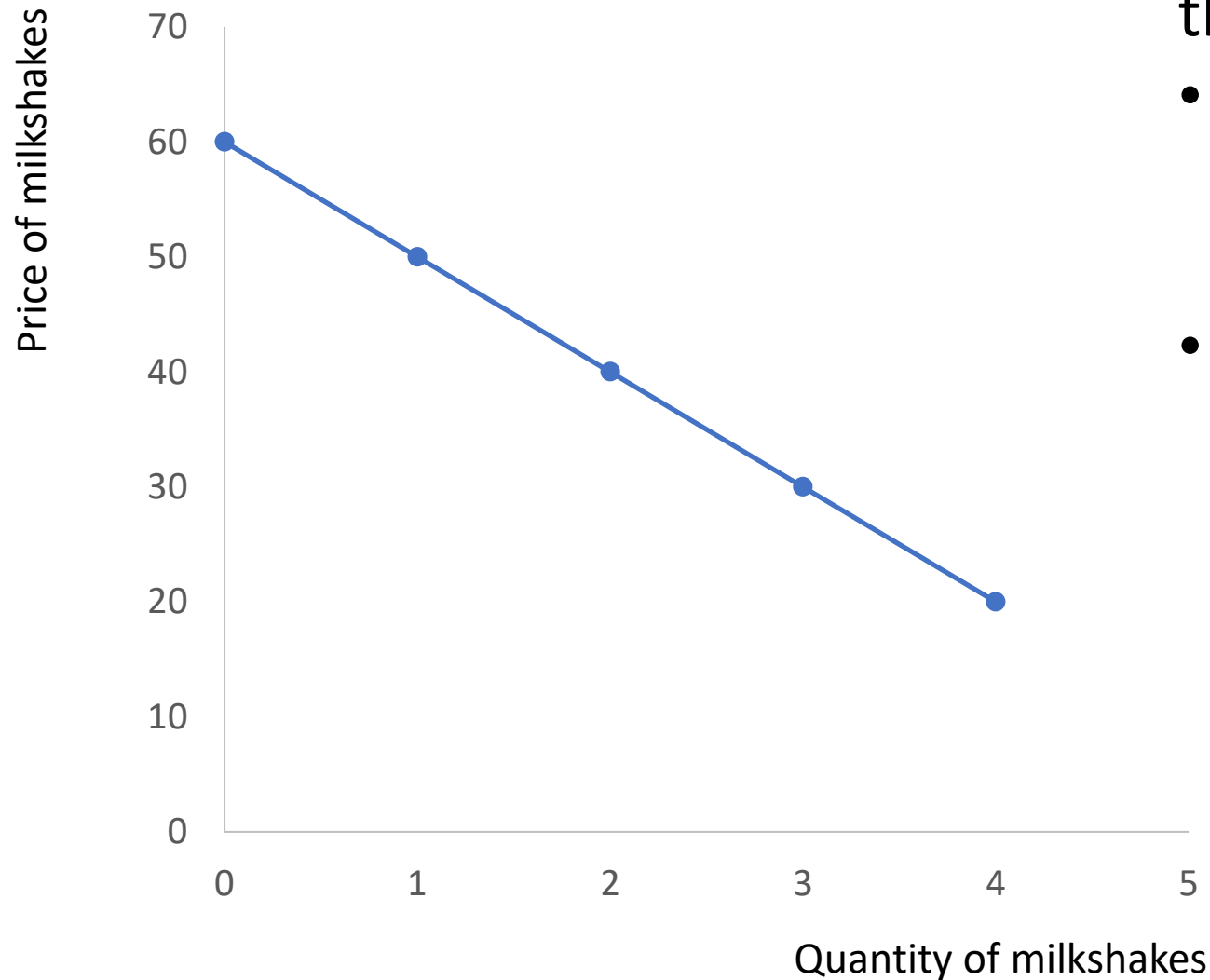
The law of demand – why?

1. Willingness to pay (WTP) decreases as quantity acquired increases: People get less utility from another unit of a product if they're already consuming a lot of it.
2. If there are good substitutes available, price increases result in some consumers rather buying from a competitor (loss of market share)
3. Especially if the product is expensive, price increases may mean people are less able to afford the product (income effect)
4. If the product is indivisible (something large like a car) more people may decide not to buy it at all when the price rises

Individual demand

- Consider a consumer who likes milkshakes
 - They derive utility from consuming milkshakes
- The consumer faces diminishing marginal utility
 - Each additional milkshake consumed results in a smaller increment to total utility than did the previous milkshake consumed
- The consumer is:
 - Willing to pay R50 for the first milkshake
 - Willing to pay R40 for the second milkshake
 - Willing to pay R30 for the third milkshake
 - Willing to pay R20 for the fourth milkshake





The **WTP** of the consumer gives us their **demand curve**:

- If the price of a milkshake is R50, the consumer would buy 1 milkshake
- If the price of a milkshake is R40, the consumer would buy 2 milkshakes
.... etc.

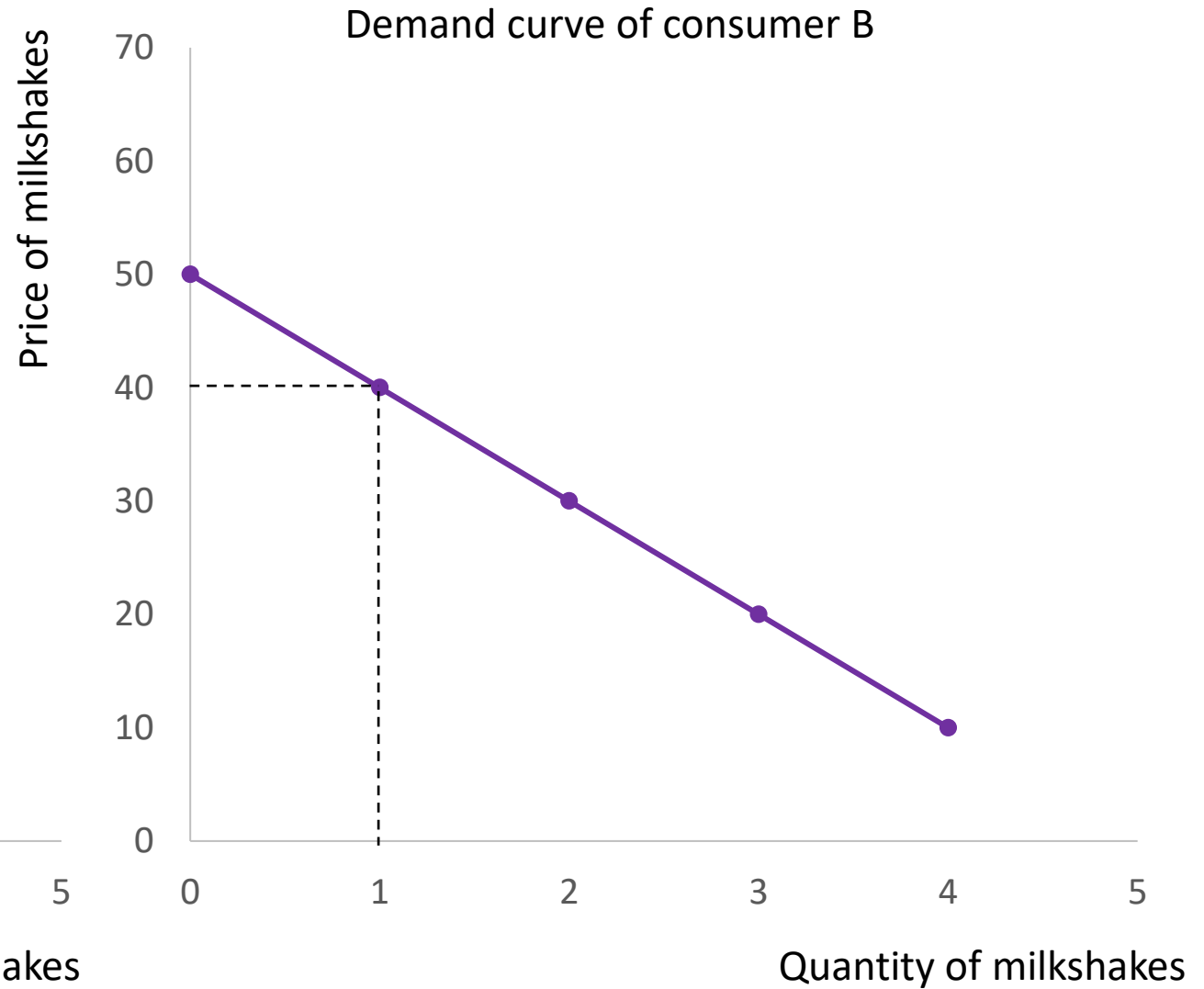
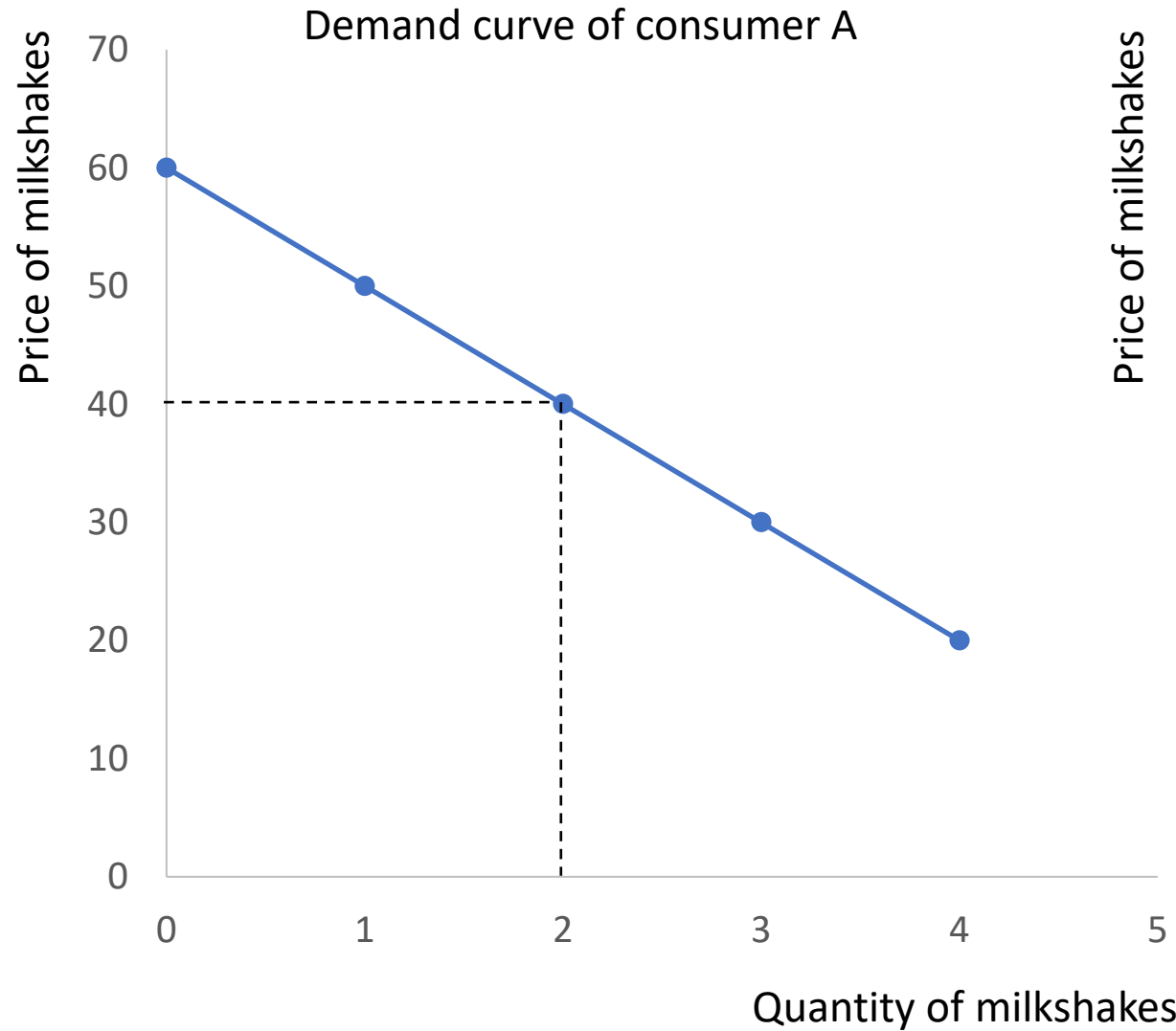
(for simplicity we have chosen to use a linear demand curve)

More than one consumer

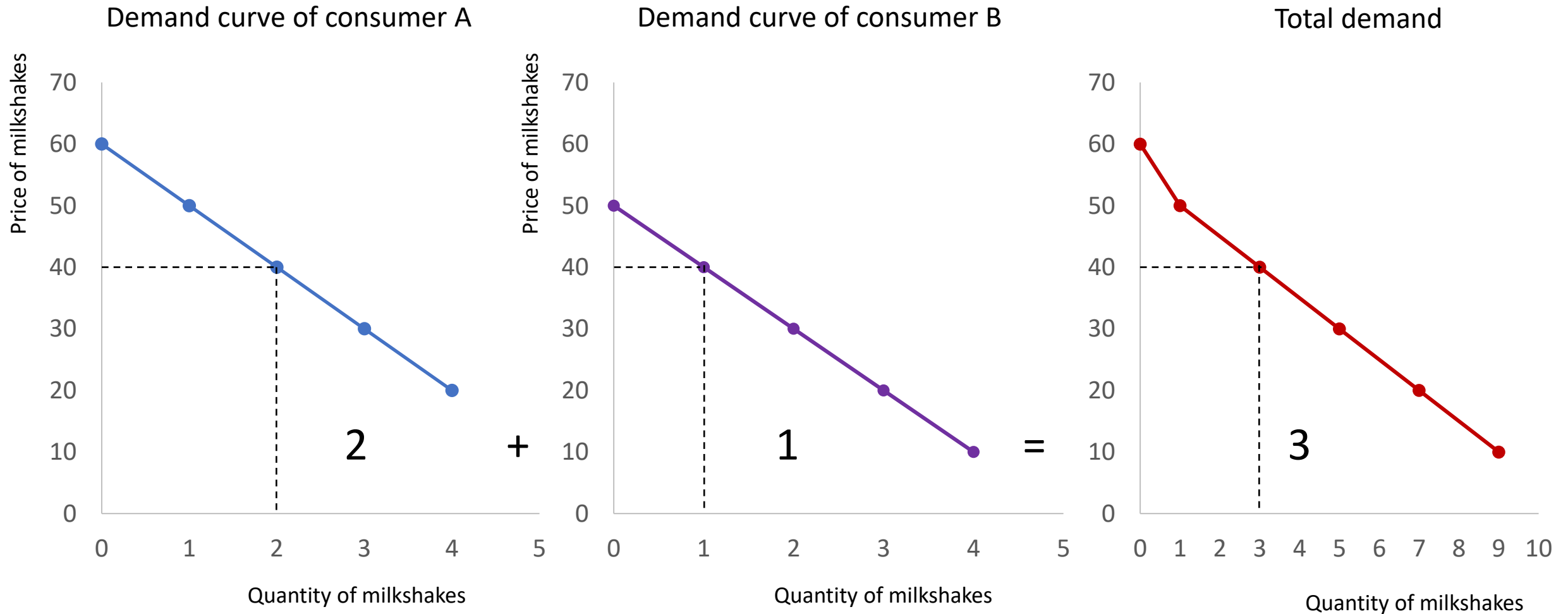
- Let's now introduce another consumer who also likes milkshakes
- This consumer (also facing diminishing marginal utility) is:
 - Willing to pay R40 for the first milkshake
 - Willing to pay R30 for the second milkshake
 - Willing to pay R20 for the third milkshake
 - Willing to pay R10 for the fourth milkshake



How many milkshakes would be bought and sold if the price was R40?

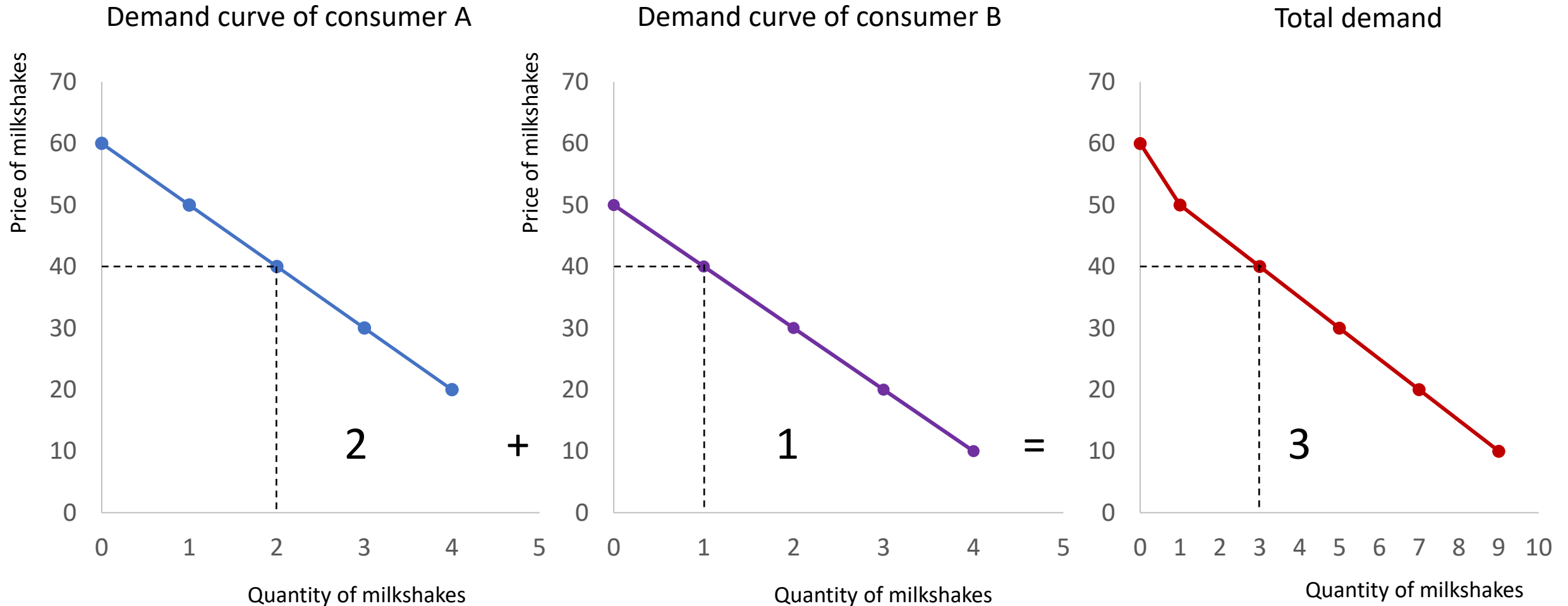


Individual and total demand



Individual and total demand

What is the combined consumers' WTP when total quantity is 5 milkshakes?



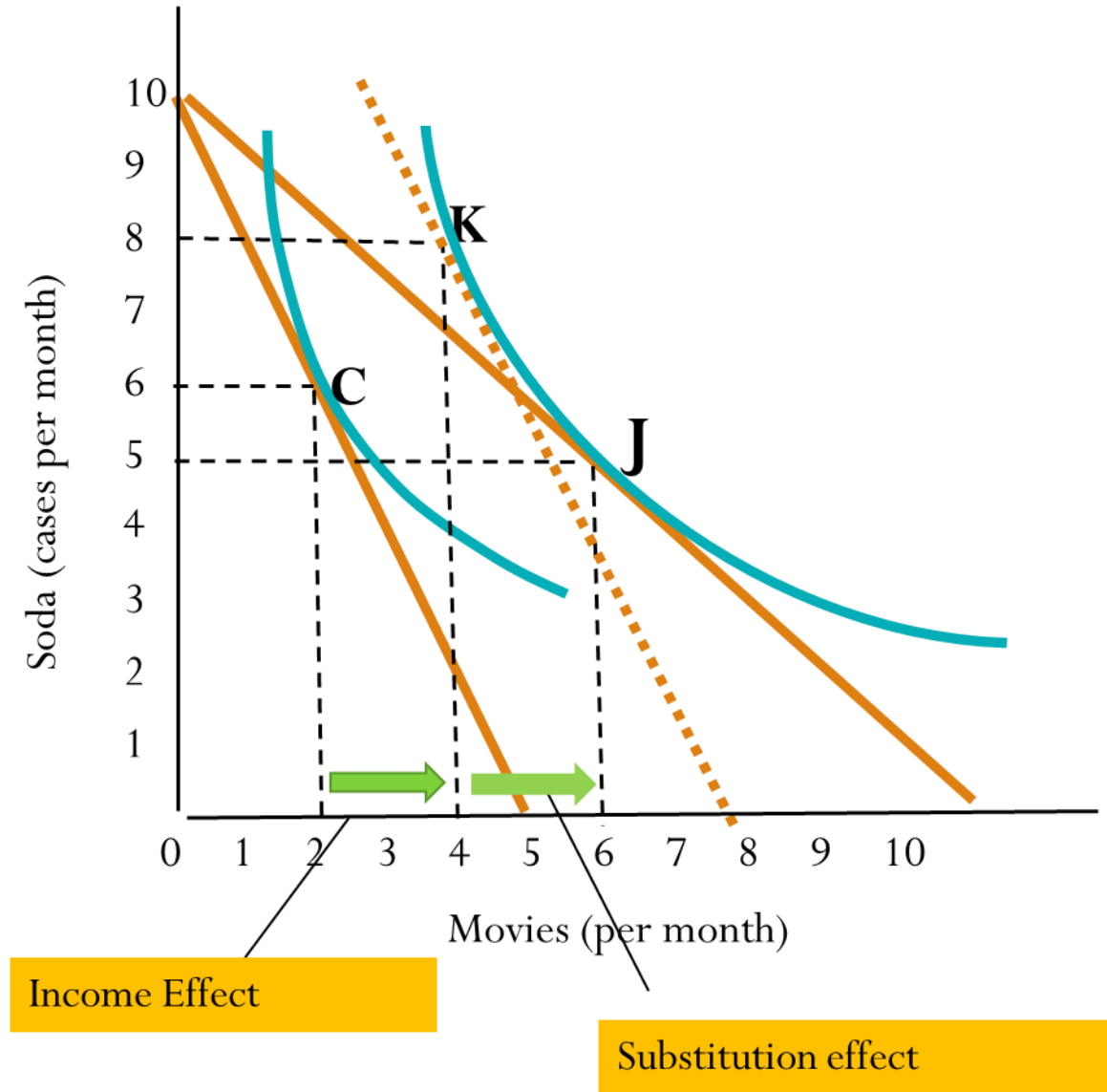
The law of demand – from consumer optimization

At the level of an individual consumer:

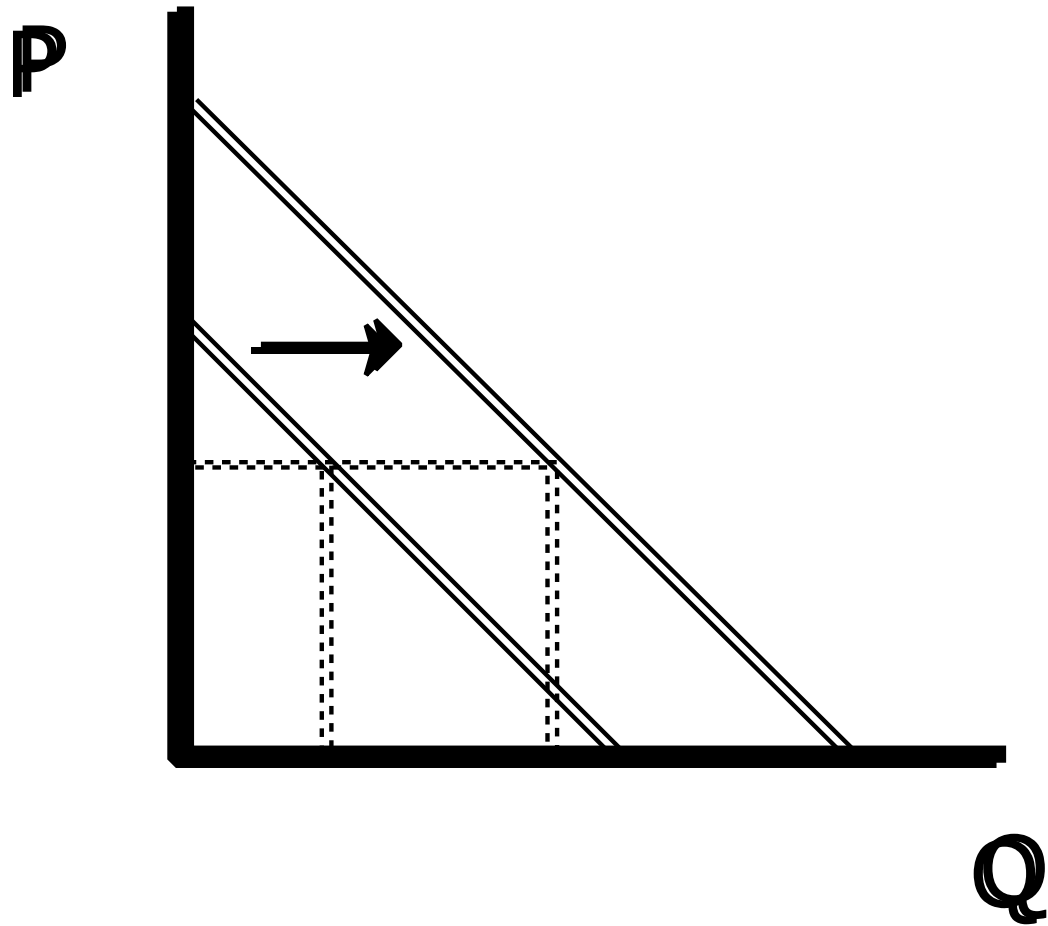
- Price of movies decreases
 - Income effect: more movies (movies are a normal good)
 - Substitution effect: more movies
 - Price (total) effect: more movies

$P \downarrow$ resulted in $Q \uparrow$

The law of demand is a standard result of the consumer optimization model

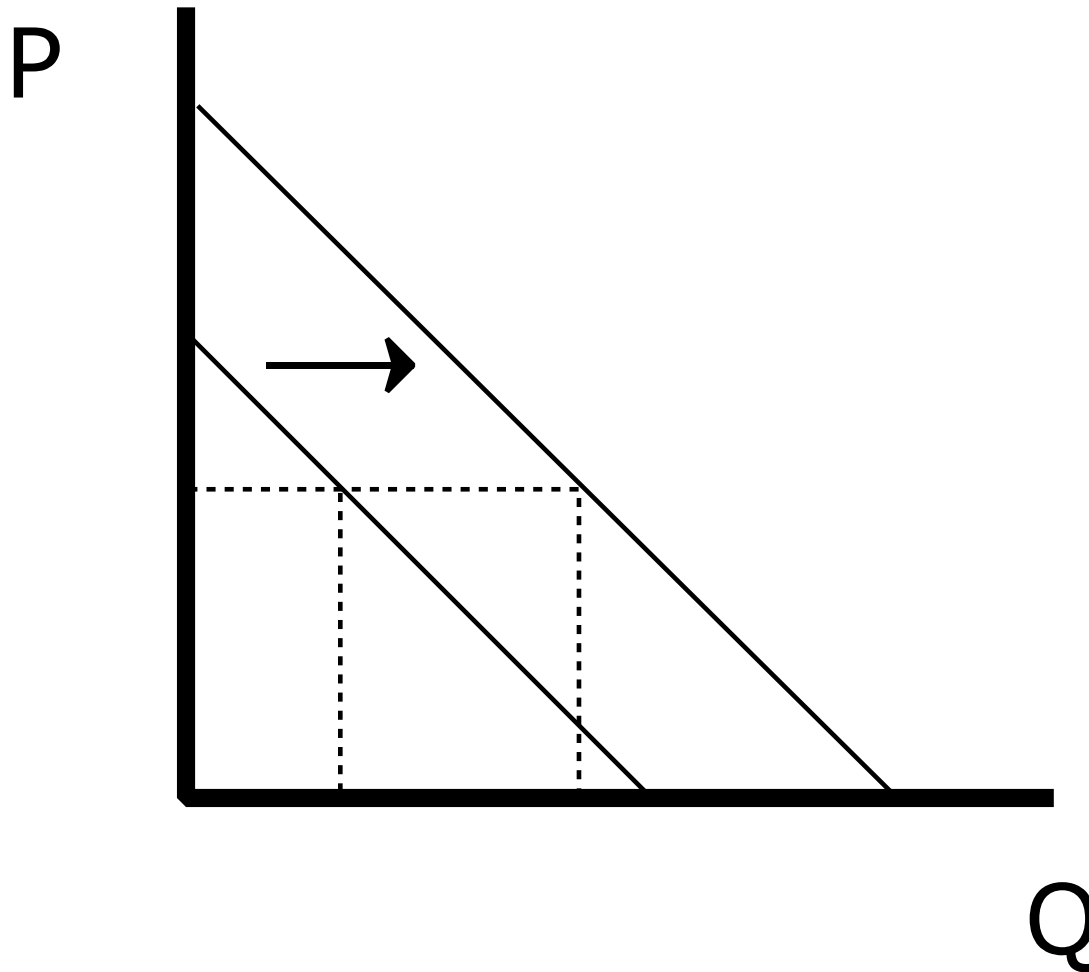


Change in Demand vs Change in Quantity Demanded



- If the **price** changes, the demand curve shows what the effect on **quantity demanded** is (a move along the demand curve).
- But if the entire demand curve shifts to the right, consumers buy a greater quantity than before at the **same** price (this is true for any given price).
- What factors can shift a demand curve?

Factors that can **change** demand



- Changes in consumer preferences
- Changes in consumer incomes
- Access to new markets (more potential consumers)
- Changes in perceived quality of the product, new features --> product innovation and advertising
- Changes in competitor's prices and perceived product quality / characteristics
- **Remember: NOT** price increase / decrease – why?

Exercise

Three consumers have demand as indicated in the table.

1. What is total quantity demanded when price is R3?
2. What is Consumer B's WTP when she consumes 50 units?
3. What is all three consumers' combined WTP (for the last unit consumed) when 50 units in total are consumed?

Price	Consumer A	Consumer B	Consumer C
1	40	50	13
2	30	46	9
3	25	43	6
4	20	40	3
5	15	35	0
6	10	30	0
7	5	25	0
8	0	20	0
9	0	15	0
10	0	10	0

Price setting firms choose P and Q combinations to maximise profit.
(we assume fixed cost per unit)

Recap

Profit:

$$\pi = TR - TC$$

Feasible price and quantity combinations are given by demand curve.

Law of Demand!

What changes quantity demanded?

What changes demand?

7.3 Production: Cost functions

- So far costs depended on quantity produced alone (we assumed that the per-unit case was constant).
- What happens when the per-unit costs vary with the level of production?
- Firms would have to consider the total costs of production that may include:
 - Rent
 - Capital (and the opportunity cost of capital)
 - Raw material
 - Wages
- Some inputs are fixed over the short run – so the firm has to pay for them even when its quantity is zero.

7.4 Cost functions

- The cost of producing any quantity of goods is given by the function

$$TC = TC(Q) = f(Q)$$

- The *average cost* is given by the function

$$AC = TC(Q)/Q$$

- The marginal cost is given by the function:

$$MC = dTC/dQ \text{ or } \Delta TC/\Delta Q$$

Note: the CORE text uses C to represent (total) cost; these slides use TC instead of C .

Exercise: constant cost per unit

Demand for this firm's chairs at different prices

Q	0	100	200	300	400	500	600	700	800	900	1000
P	R300	R270	R240	R210	R180	R150	R120	R90	R60	R30	R0
TC											
AC											

The firm's unit cost of production for a chair is R60. Calculate at each Q:

1. TC
2. AC
3. MC



Exercise: fixed cost per unit

Demand for this firm's chairs at different prices

Q	0	100	200	300	400	500	600	700	800	900	1000
P	R300	R270	R240	R210	R180	R150	R120	R90	R60	R30	R0
TC	R0	$100 \times \text{R}60 = \text{R}6000$	$200 \times \text{R}60 = \text{R}12000$								
AC	- (undefined)	$\text{R}6000/100 = \text{R}60$	$\text{R}12000/200 = \text{R}60$								
MC	- (undefined)	$\text{R}12000/200 = \text{R}60$									

The firm's unit cost of production for a chair is R60. Calculate at each Q:

1. TC
2. AC
3. MC

The first few are done for you.

Using the **midpoint method**:

Q = 100 is midway between Q = 0 and Q = 200; therefore, we approximate MC using $dTC/dQ = (\text{R}12000 - \text{R}0)/(200-0) = \text{R}60$

7.4 Cost curves: Varying per-unit costs

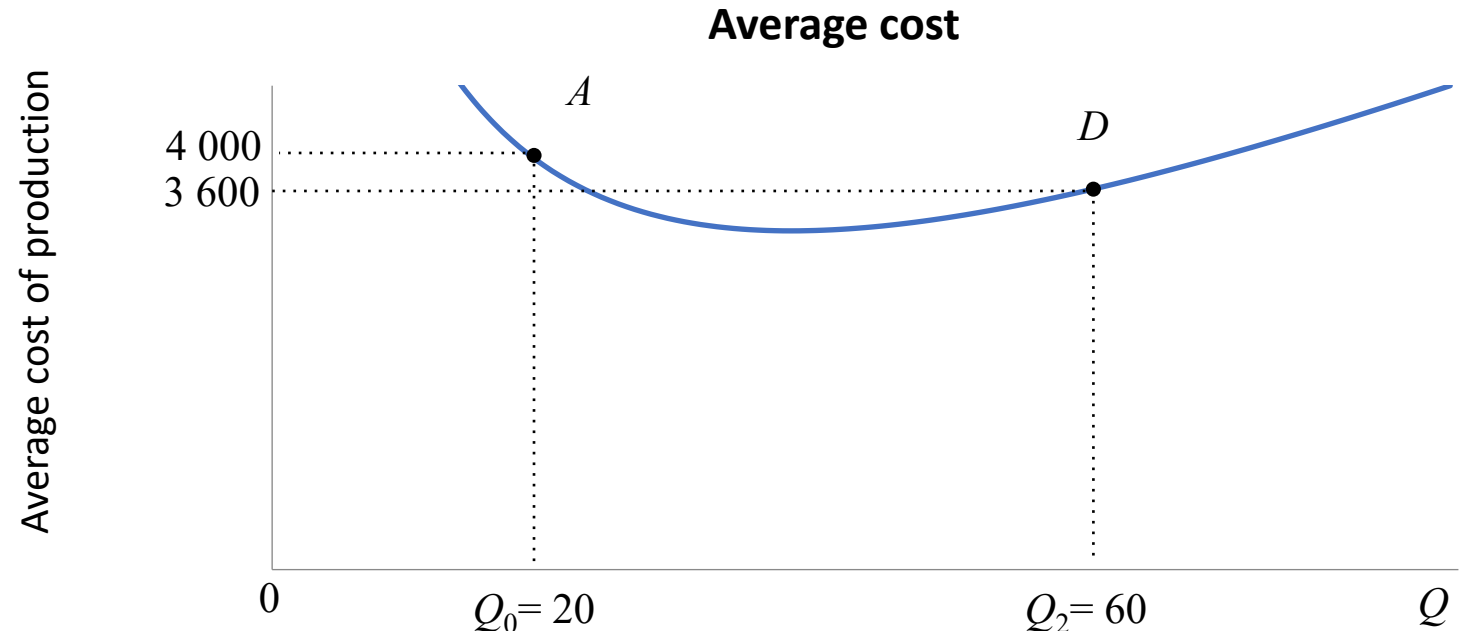
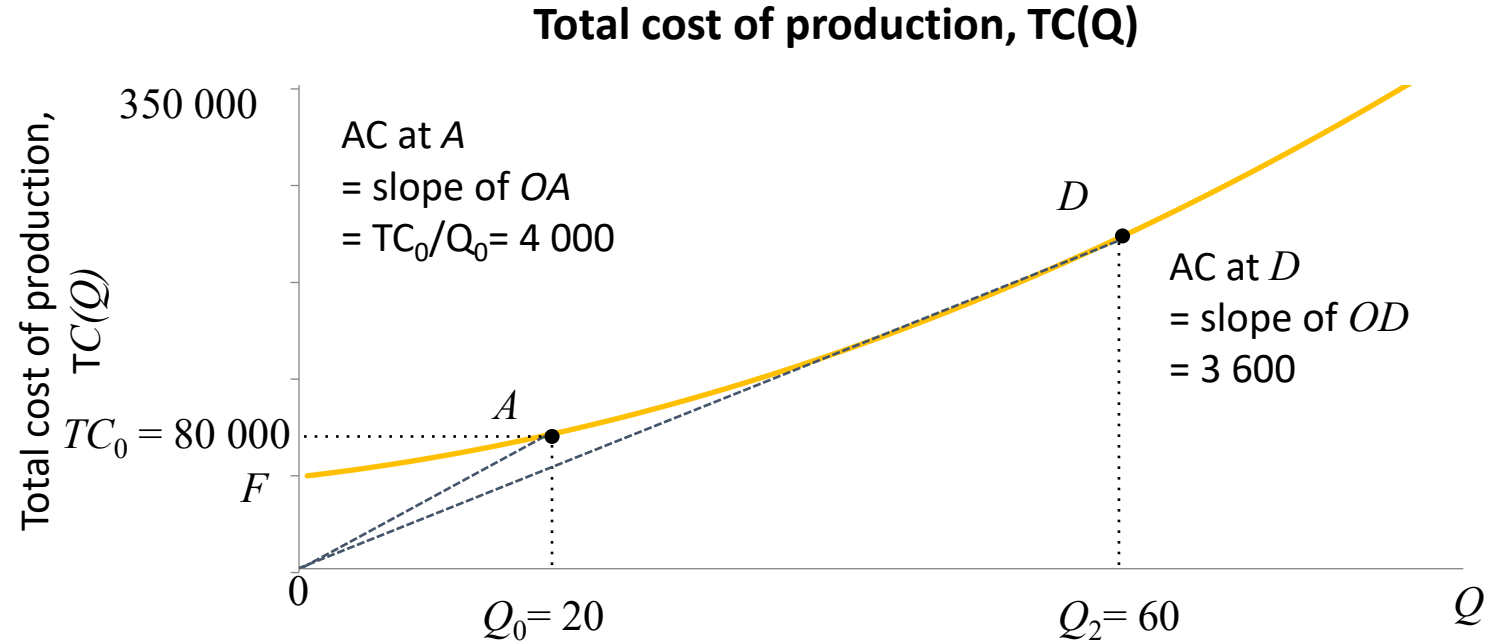
Total Costs (TC), Average Costs (AC) and Marginal Cost (MC) of production of Beautiful Cars				
Q	0	20	40	60
$TC = f(Q)$	48 000	80 000	136 000	216 000
$AC = TC/Q$	undefined	4 000	3 400	3 600
$MC = dTC/dQ$	undefined	2 200	3 400	4 600

Make sure you understand how we come out to these values.

Question: What is this firm's **fixed cost**?

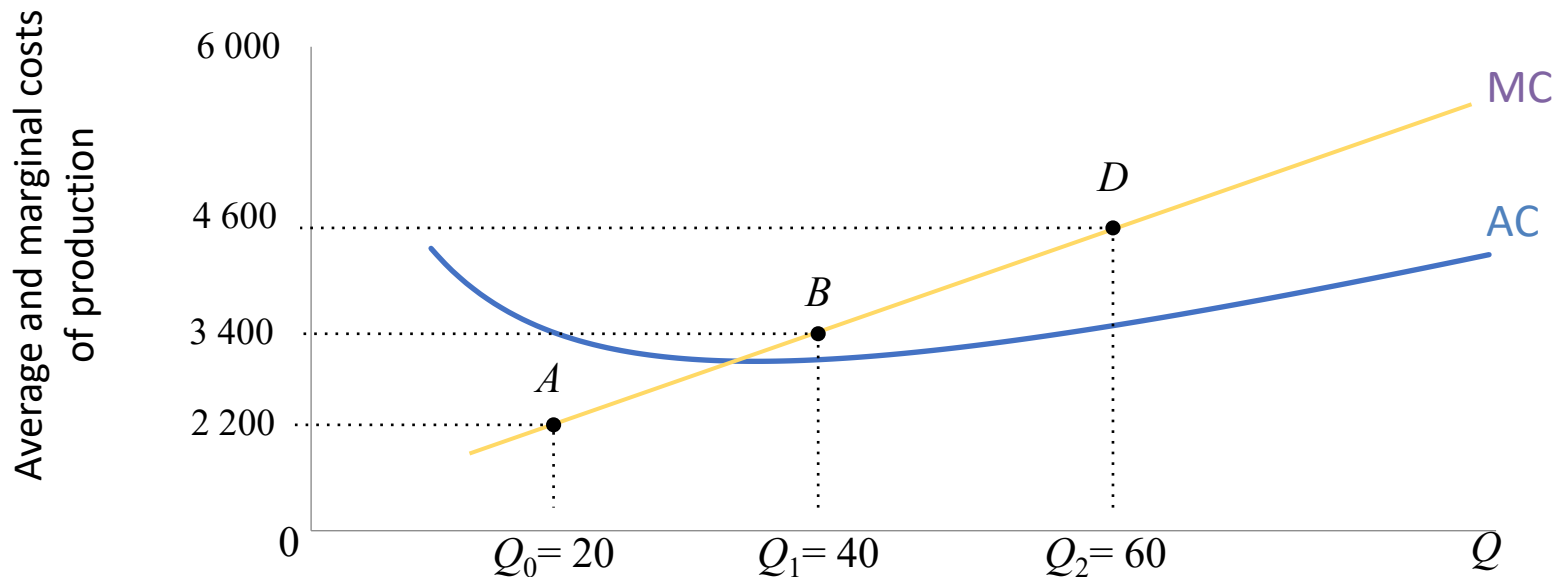
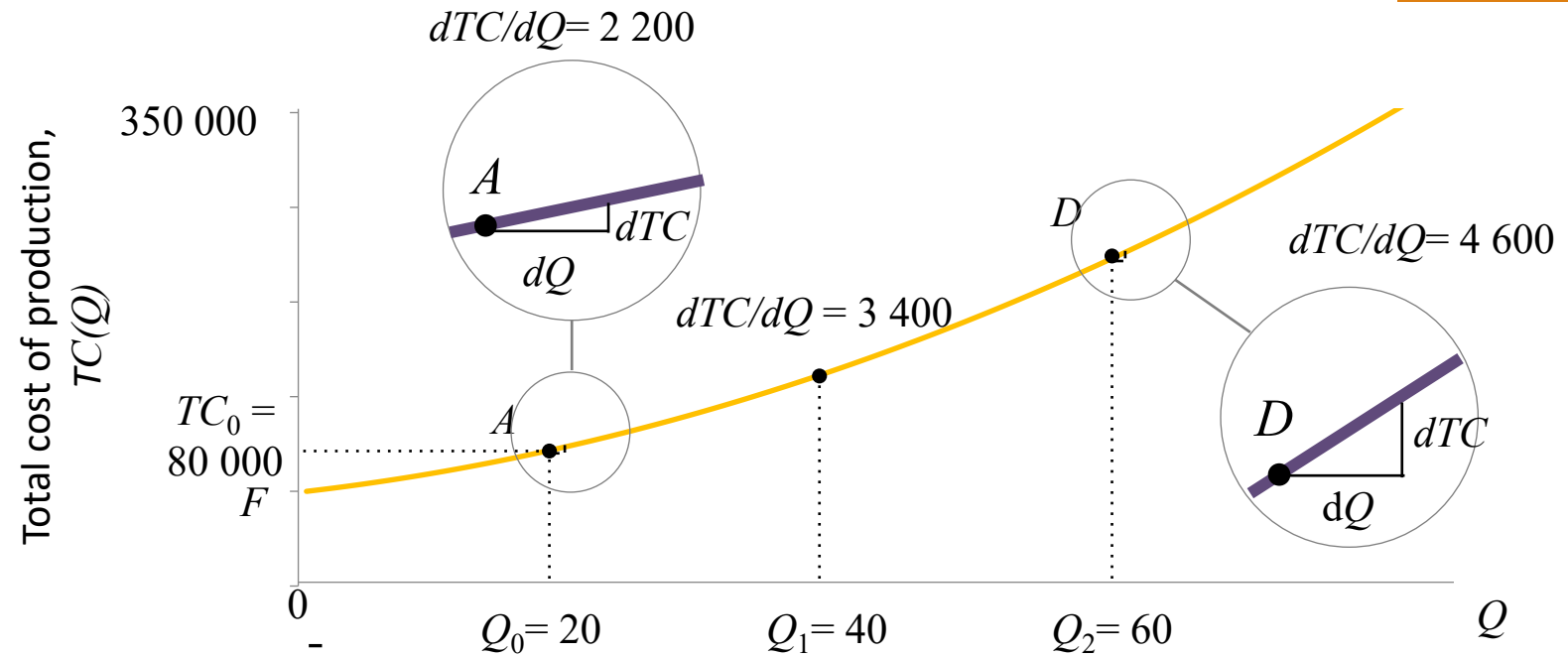
7.4 Cost curves

- The firm's cost function shows how TC increases when Q increases.
- The AC curve also changes with Q
 - Point A:** When Q is low, AC is decreasing and slopes downwards
 - Point D:** When Q is high, AC is increasing and slopes upwards
- The AC curve is U-shaped



7.3 Cost curves

- MC = slope of the cost function
- The additional cost of producing one additional unit of output
- MC intersects AC at AC 's minimum:
 - When $MC < AC$: AC is decreasing
 - When $MC > AC$: AC is increasing



Average and marginal cost curve shapes

- Average cost is often U-shaped
 - At low quantities, average cost falls, and at high quantities, average cost rises.
- Why?

Average and marginal cost curve shapes

- Firms usually have some **fixed inputs** that do not vary as quantity is changed (this also explains **fixed costs**)
- Examples: buildings and other large capital installations that cannot easily be changed
- In the **long run** all inputs can be changed, but quantity decisions must be made on shorter timescales, so only some inputs are increased to increase quantity
- At low quantities, there could be insufficient utilization of fixed inputs, and few opportunities for cost-saving specialization, so average cost is high
- Average cost also falls because fixed costs are divided by increasing number of units of output
- At high quantities, there could be overutilization of fixed inputs, leading to holdups, overcrowding, management problems or frequent breakdowns, and thus higher average cost
- Recall: Diminishing average product of labour (Unit 2)

Average and marginal cost curve shapes

- Marginal cost is sometimes shown as U-shaped (for similar reasons given in previous slides for AC), but often just as an upwards-sloping curve (for simplicity)

Cost functions (mathematical)

- Suppose that the TC function for a firm is given by:

$$TC(Q) = 1100 + 140Q + 0.2Q^2$$

- The AC function will be given by:

$$AC = TC/Q = 1100/Q + 140 + 0.2Q$$

- The MC function will be given by:

$$MC = dTC/dQ = 140 + 0.4Q$$

Exercise: Cost functions

- Given the TC function:

$$TC(Q) = 62\,500 + 500Q + Q^2$$

- What is the marginal cost at a production level (Q) of 1050?
- At what Q is AC at a minimum?

Back to profit maximisation...

- Earlier we looked at maximising profit by using $\pi = TR - TC$ and identify the price-quantity combination on the demand curve that yields the highest profit.
- After taking a more in depth look at costs, we now introduce a second way in which we can calculate the profit maximising price and quantity.
- We do this by using the concepts of marginal revenue (MR) and marginal cost (MC).

7.6 Price setting using marginal revenue

- An alternative way of maximizing profit relies on a new concept, **marginal revenue (MR)**.
 1. The change in revenue (TR) when the quantity (Q) increases by 1.
 - $MR = \frac{\Delta TR}{\Delta Q}$ when $\Delta Q = 1$
- OR
2. The rate of change of TR as Q changes (slope of the TR function)
 - $MR = \frac{dTR}{dQ}$

The two definitions give approximately the same result in most cases and we use both definitions depending on the context.

7.6 Price setting using marginal revenue*

(* Note that we changed the values in Fig 7.12a to make it consistent with Fig 7.12b)

$$Q = 19$$

$$P = \$6\,480$$

$$TR = \$123\,120$$

TR = $P \times Q$, the **area** of the rectangle below the (P, Q) coordinate

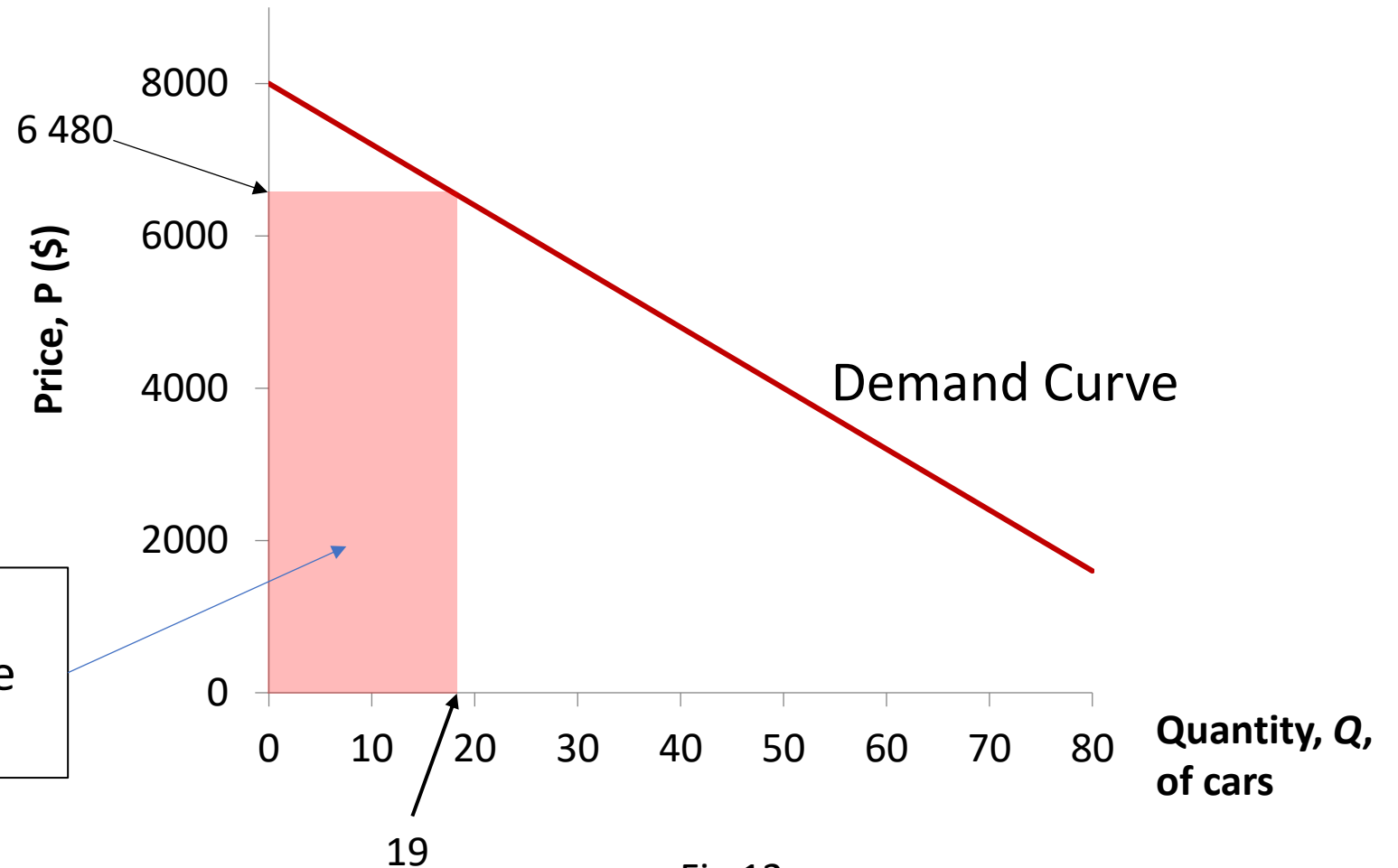


Fig 12a

7.6 Price setting using marginal revenue*

(* Note that we changed the values in Fig 7.12a to make it consistent with Fig 7.12b)

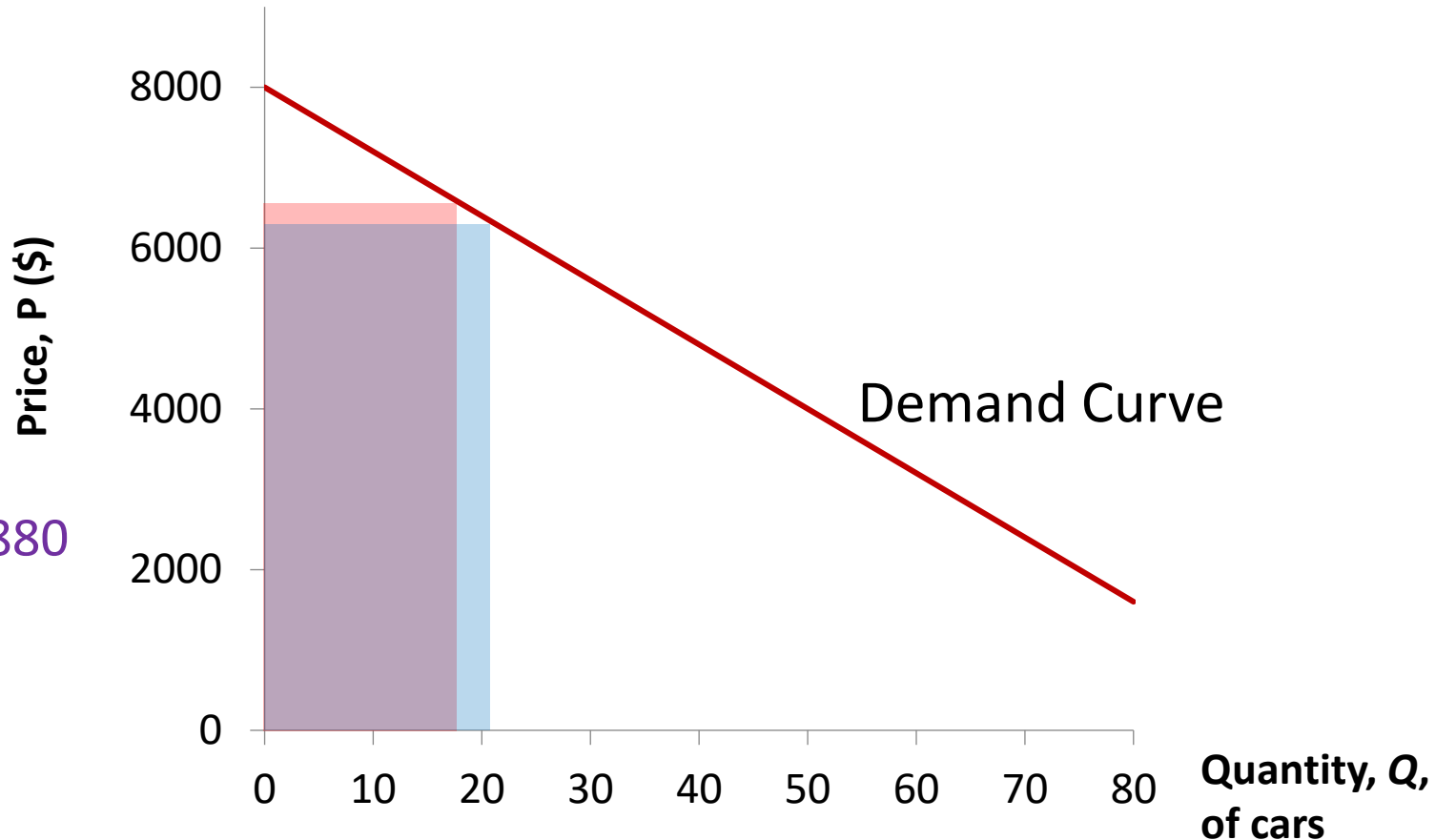
$$Q = 20$$

$$P = \$6\,400$$

$$TR = \$128\,000$$

Gain in TR : $\$128\,000 - \$123\,120 = \$4\,880$

Thus MR = \$4 880



7.6 Price setting using marginal revenue

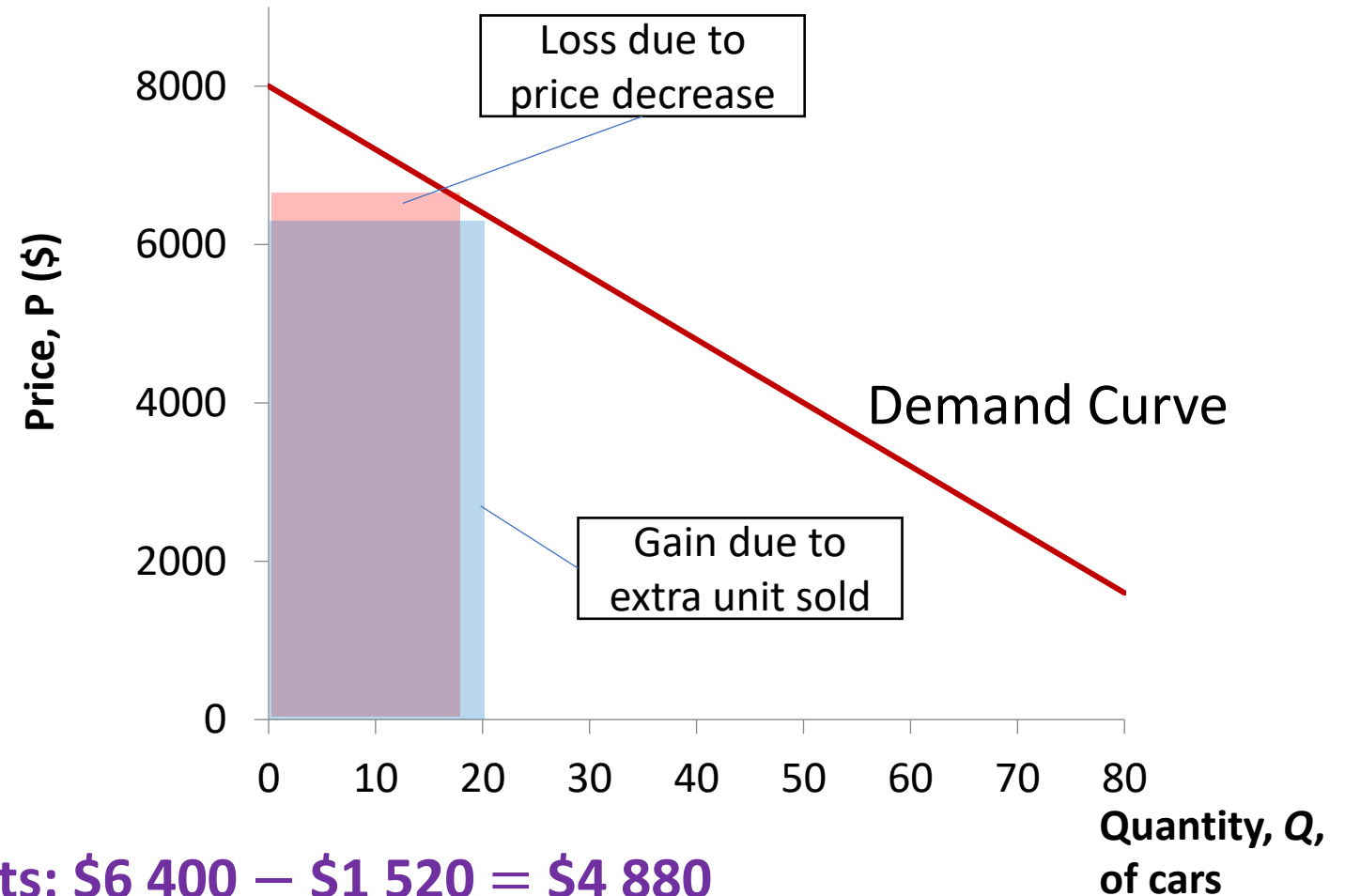
TWO things happen to TR
when Q is increased by 1:

1) 20th car is sold at \$6 400.
Gain in TR of $1 \times \$6\,400$ due to
20th car sold

2) The other 19 cars used to be
sold for \$6 480; now they are
sold for \$80 less.

Loss in TR of $19 \times \$80 = \$1\,520$

MR is the sum of these two effects: $\$6\,400 - \$1\,520 = \$4\,880$



Using calculus:

$$TR = P \times Q$$

$$MR = \frac{dTR}{dQ} = \frac{dP}{dQ} \times Q + P \times \frac{dQ}{dQ}$$

(using the product rule of differentiation)

$$\frac{dQ}{dQ} = 1, \text{ so,}$$

$$MR = \boxed{\frac{dP}{dQ} Q} + \boxed{P}$$

Loss due to
price decrease

Gain due to
extra unit sold

$\frac{dP}{dQ}$ is the slope of demand curve – it measures how fast prices drop when quantity is increased

$$\frac{dP}{dQ} < 0, \text{ so}$$

$$MR < P$$

For a linear demand curve..

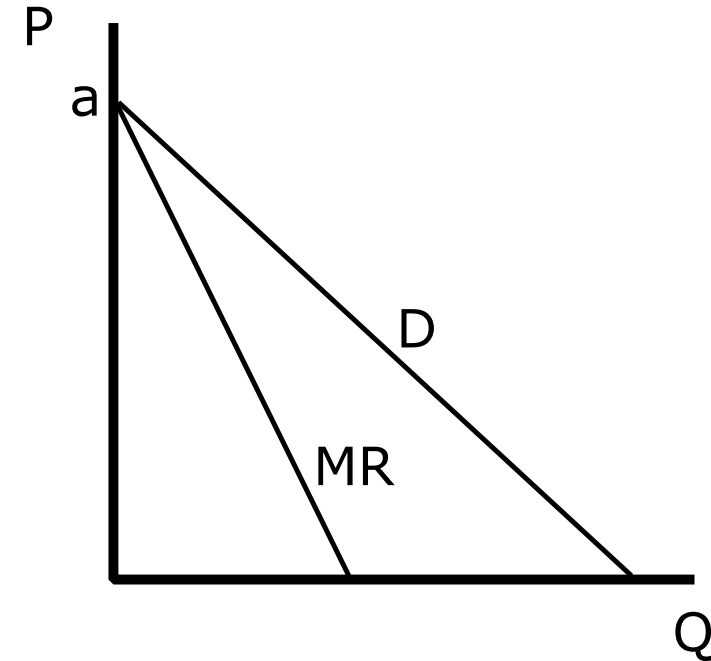
$$P = a - bQ$$

$$TR = P * Q$$

$$TR = aQ - bQ^2$$

$$\mathbf{MR = a - 2bQ}$$

(derived from TR)



MR's vertical intercept (a) is the same as the demand curve's

MR's slope (-2b) is twice the slope of demand curve (-b)

The MR curve lies below the demand curve ($MR < P$)

Marginal revenue

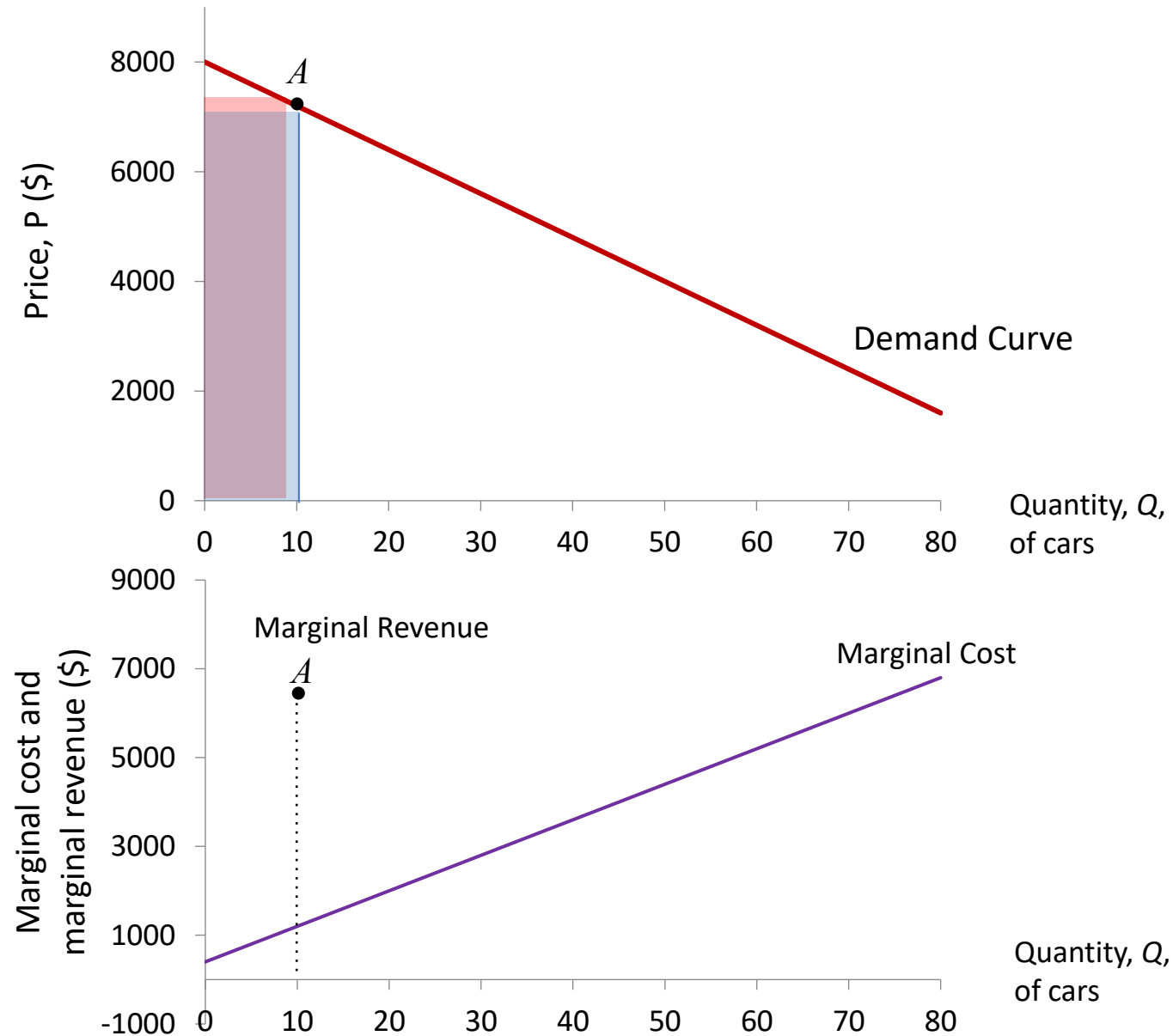
Demand function: $P = 8\,000 - 80Q$

If $Q = 9$: $P = \$7\,280$ and $TR = \$65\,520$

If $Q = 10$: $P = \$7\,200$ and $TR = \$72\,000$

So, MR of selling the:

10th unit is \$6 480



Marginal revenue

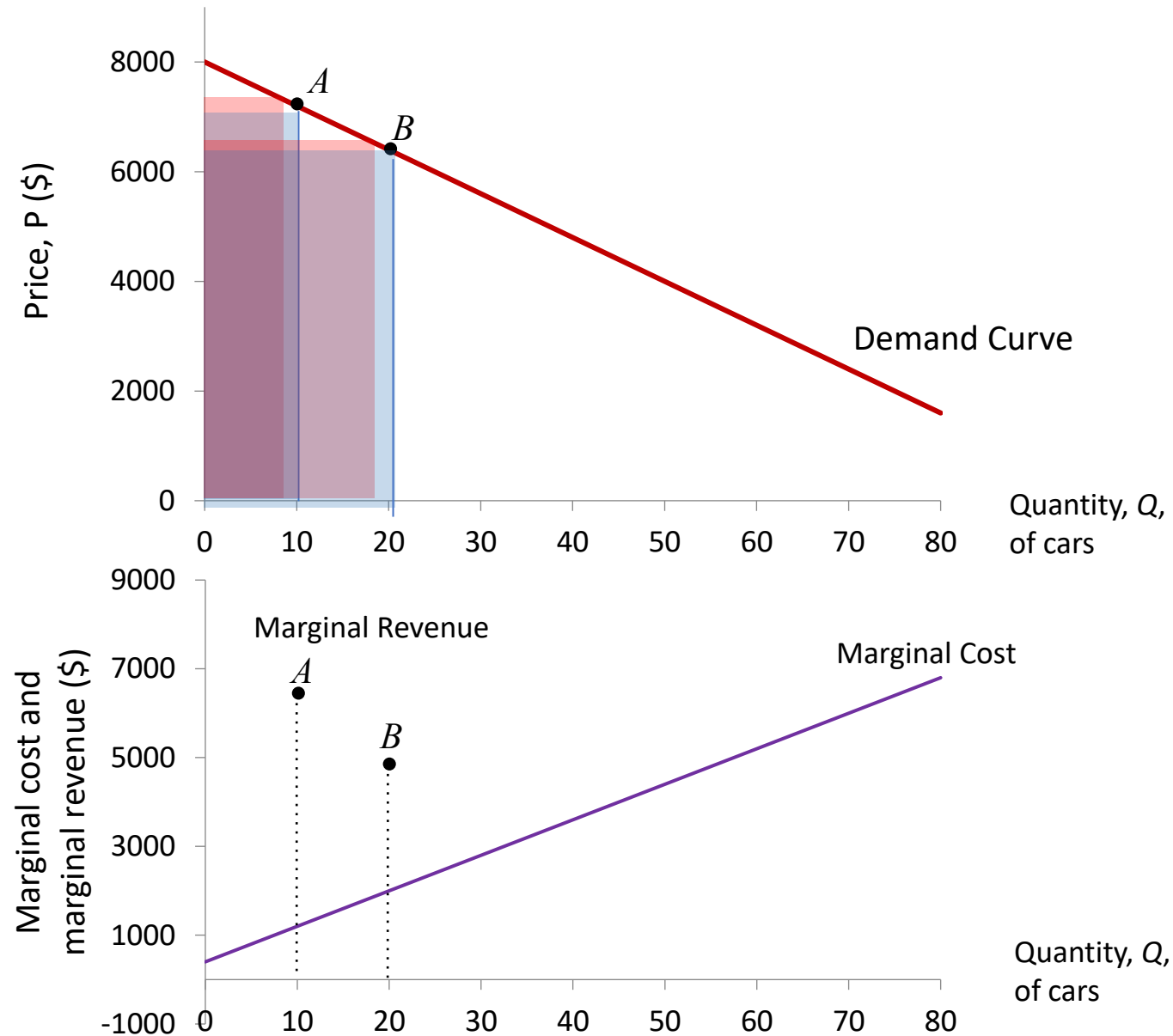
Demand function: $P = 8\,000 - 80Q$

If $Q = 19$: $P = \$6\,480$ and $TR = \$123\,100$

If $Q = 20$: $P = \$6\,400$ and $TR = \$128\,000$

So, MR of selling the:

20th unit is \$4 880



Marginal revenue

MR of selling the:

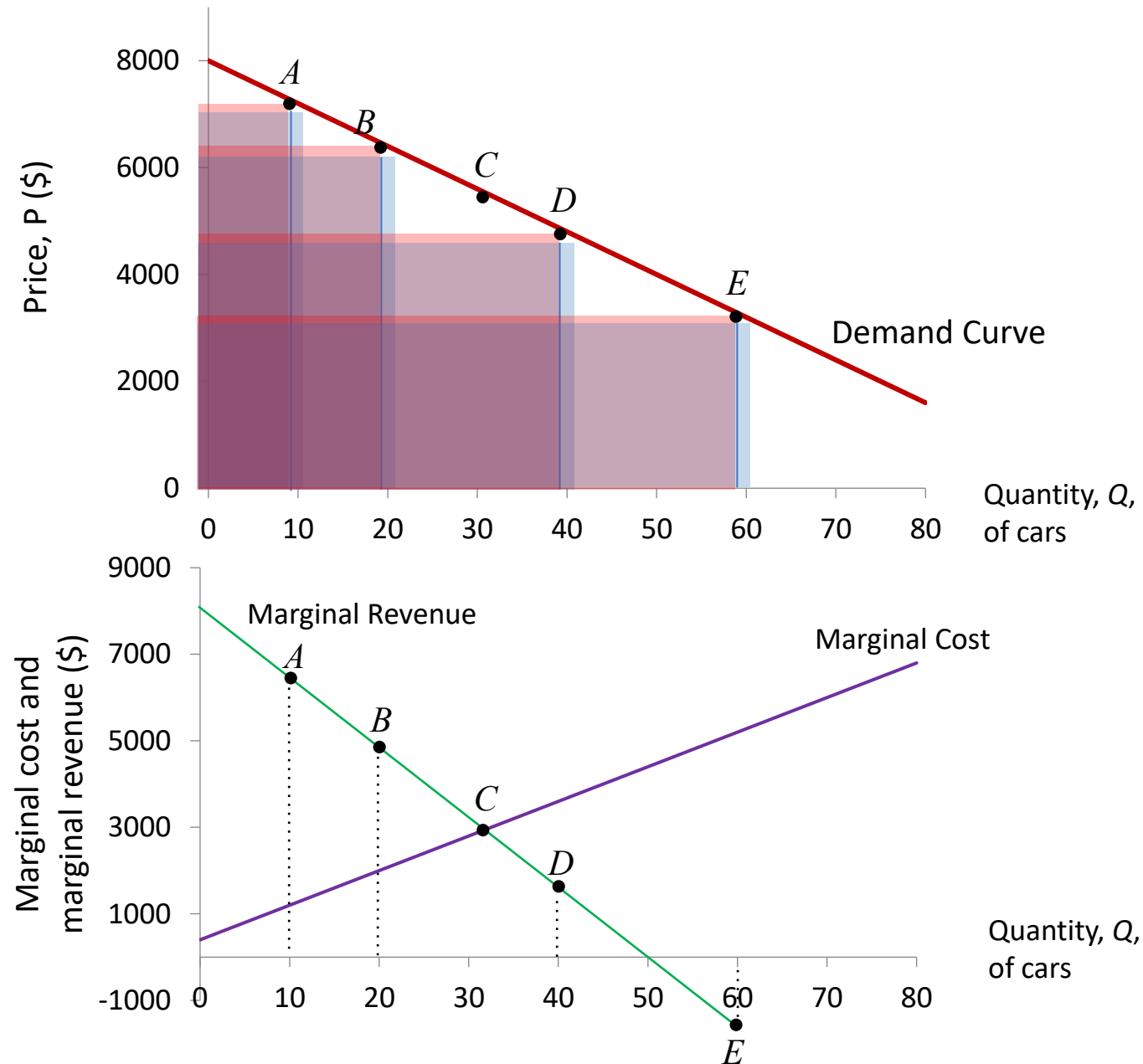
10th unit is \$6 480

20st unit is \$4 880

40st unit is \$1 680

60st unit is -\$1 520

(MR will be linear if the demand curve is linear)

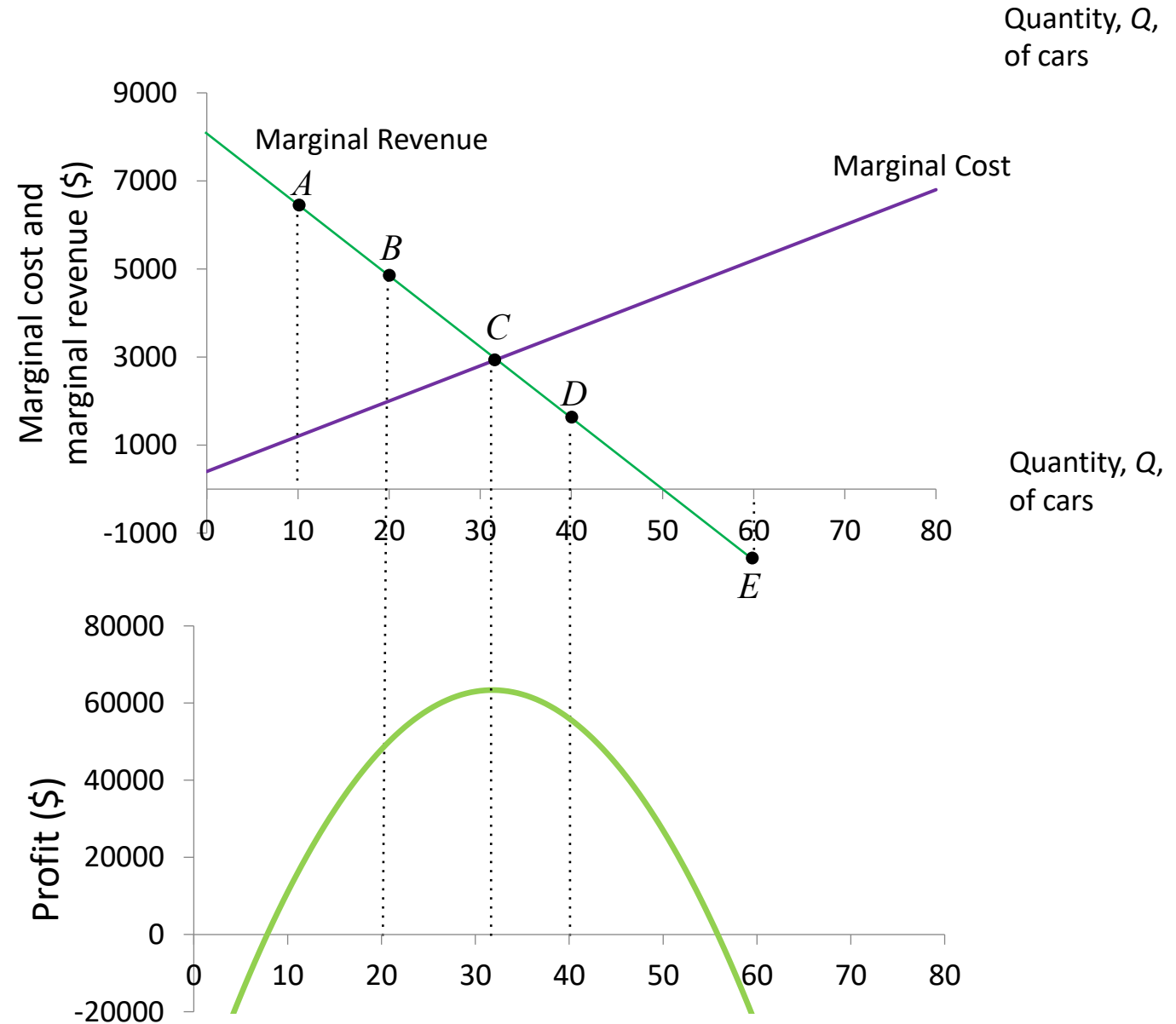


Marginal revenue

If $MR > MC$ (e.g. B), an extra unit would increase π so produce more

If $MR < MC$ (e.g. D), one unit less would increase π so reduce production

Max profit: $MR = MC$
(at C, where $Q = 32$)



Marginal revenue and profit maximization (using calculus)

$$\text{Profit } (\pi) = TR - TC$$

$$\pi \text{ is maximized when } \frac{d\pi}{dQ} = 0$$

$$\begin{aligned}\frac{d\pi}{dQ} &= \frac{dTR}{dQ} - \frac{dTC}{dQ} \\ &= MR - MC\end{aligned}$$

$$\pi \text{ is maximized when } MR - MC = 0$$

$$MR = MC$$

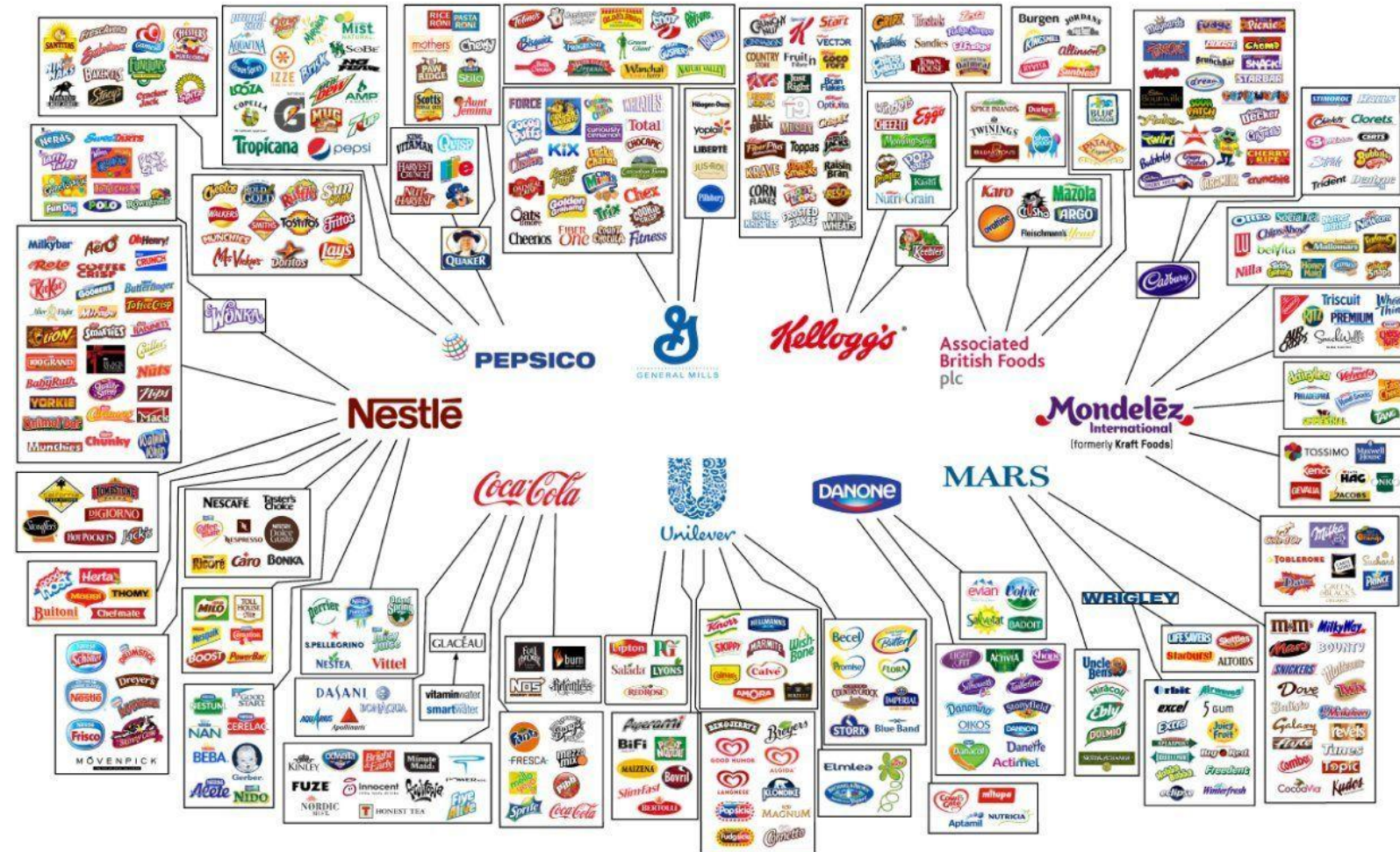
Exercise: Marginal Revenue

The demand function is $Q = 100 - \frac{1}{80}P$.

1. Give a function for total revenue (TR).
2. What are the gain (due to selling an extra unit) and loss (due to the price decrease on all the other units sold) parts of MR when Q increases from 10 to 11?
3. Derive a function for marginal revenue (MR) using calculus.
4. What is MR when $Q = 10.5$ (using the MR function)?

7.3 Economies of scale

Why do Large firms exist?



Branded goods are all (somewhat) differentiated.

7.3 Economies of scale

Economies of scale/ increasing returns to scale occur when, for example, doubling all of the inputs to a production process *more* than doubles the output.

7.3 Economies of scale

- Large firms like Walmart, Toyota, and Amazon are large due to *economies of scale* or *increasing returns*.
- Industries characterised by small firms generally have *constant returns to scale* or *decreasing returns to scale*.



Toyota benefits from technological advantages, cost advantages, and network economies.

7.3 Economies of scale

Economies of scale result from:

- *Technological advantages*: when large scale production requires fewer inputs per unit of output.
- *Cost advantages*: when large firms purchase inputs at lower cost.
- *Network economies*: when people purchase a good because it has many consumers (or users).



Toyota benefits from technological advantages, cost advantages, and network economies.

7.3 Economies of scale: technological advantages



Doubling the diameter (and circumference) of a pipe quadruples its area (and flow).



Rawhide Energy Station converts coal into energy more efficiently than you.

7.3 Economies of scale: cost advantages



Shoprite pays suppliers lower prices than Spaza shops.

7.3 Economies of scale: network economies



Market capitalization in fall 2018

- Apple: \$1.05 trillion
- Google: \$763 billion
- Facebook: \$443 billion
- Amazon: \$859 billion

You use these firms because other people use these firms.

Diseconomies of scale

Diseconomies of scale occur when, for example, doubling all of the inputs to a production process *less* than doubles the output.

Diseconomies of scale



Increasing the workforce 10-fold increases production less than 10-fold. If people can only supervise 10 workers, A firm with 10^x production workers will have x levels of management.

Setting the price of beautiful cars



Most firms have both economies of scale and diseconomies of scale.

Economies and diseconomies of scale explain firms' **long run** total, average and marginal cost relationships, i.e. when all inputs can be changed.

Product differentiation (Monopolistic competitors) vs. **Monopolies**

Firms selling differentiated products compete against other firms selling their own varieties of the same type of product

Even though such firms have market power, they still face a degree of competition for market share: if they raised their prices, they would lose some market share to their competitors

Monopolies face no competition at all:

- A single firm controls the entire supply in a market

- There are no close substitutes for its product

Firms selling differentiated products and monopolies are both price setters
Both are constrained by downwards sloping demand curves

Monopolies

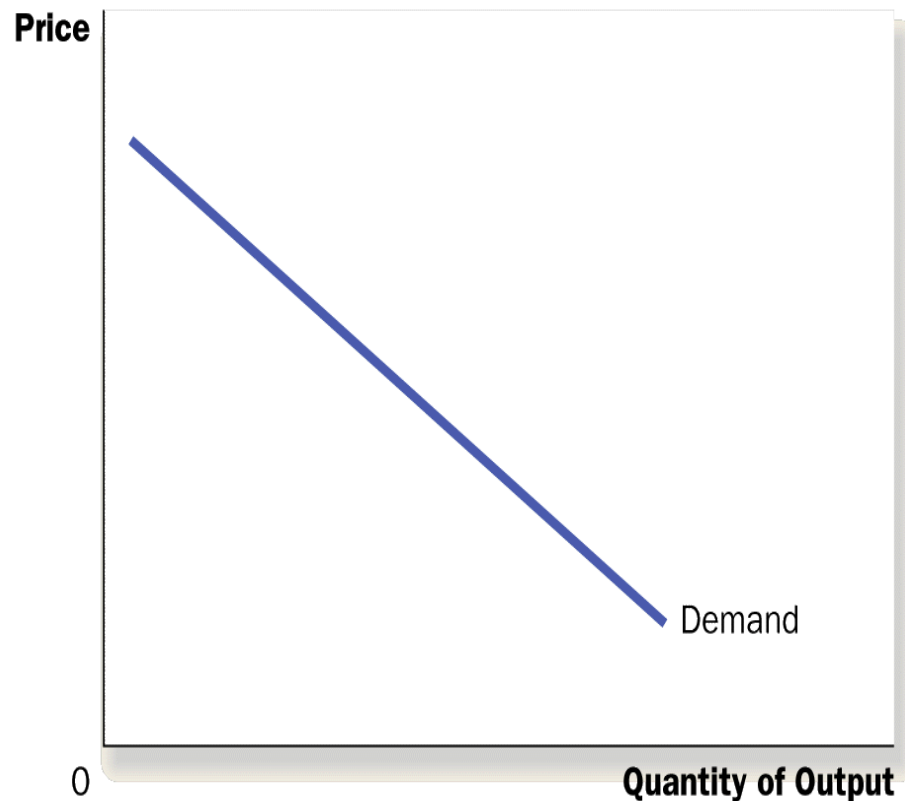
- Why do some markets have only one seller?
- How do they protect their market (barriers to entry)?
- How does a monopoly determine the quantity to produce and the price to charge?

Barriers to entry

1. Exclusive ownership of a resource - (De Beers)
2. Legal - (Telkom)
 - Patents, copyright
3. Cost of production (economies of scale) – (Eskom)
 - Natural Monopoly

Monopoly's demand = Market demand

(b) A Monopolist's Demand Curve



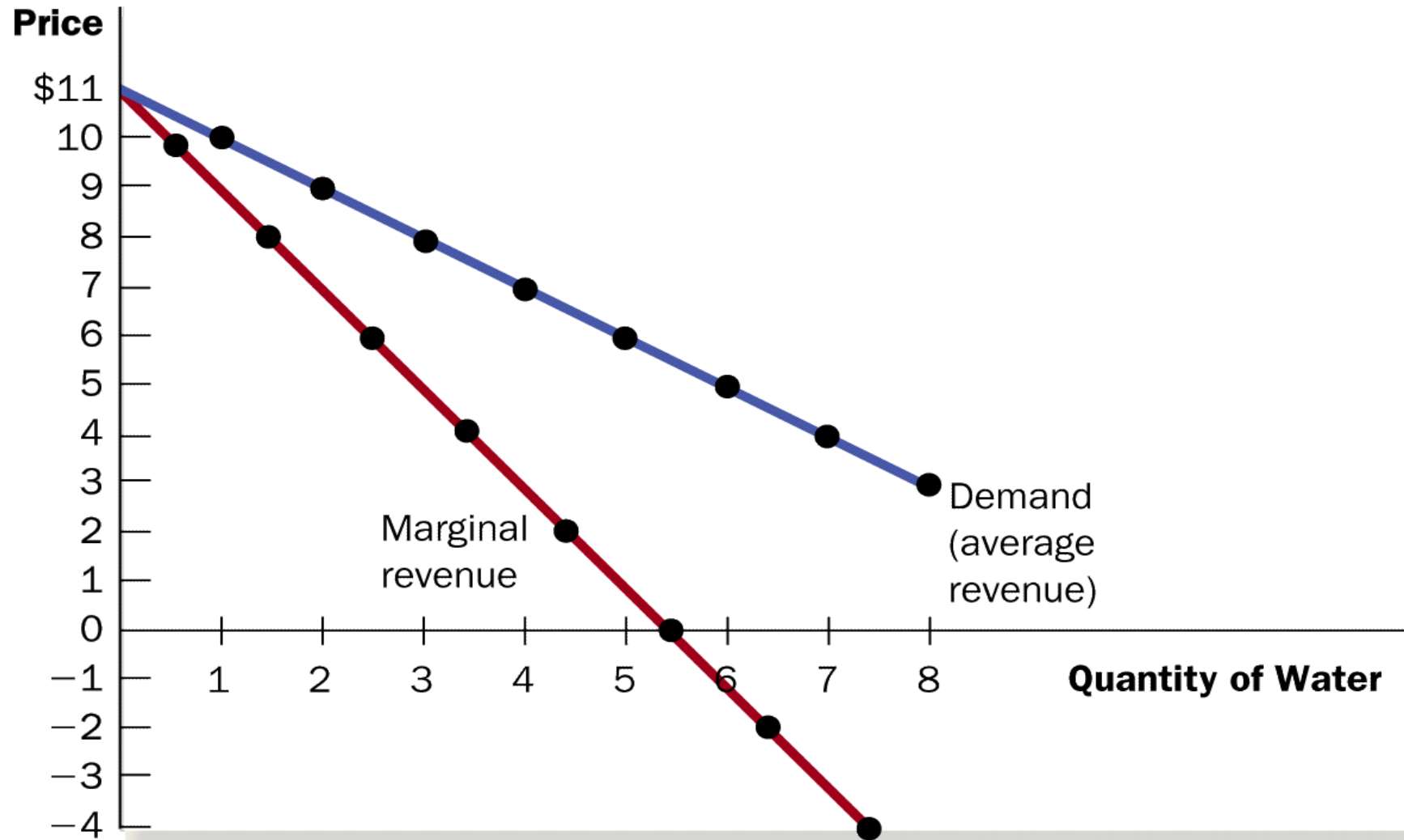
Downwards sloping demand curve: Not all P, Q combinations are feasible

Note: the graphical analysis is identical to that of firms selling differentiated products. The only difference is that it now refers to the whole market.

Monopolist's Revenue

Quantity of Water	Price	Total Revenue	Average Revenue	Marginal Revenue
(Q)	(P)	($TR = P \times Q$)	($AR = TR/Q$)	($MR = \Delta TR/\Delta Q$)
0 gallons	\$11	\$ 0	—	
1	10	10	\$10	\$10
2	9	18	9	8
3	8	24	8	6
4	7	28	7	4
5	6	30	6	2
6	5	30	5	0
7	4	28	4	−2
8	3	24	3	−4

D and MR curves of the Monopolist



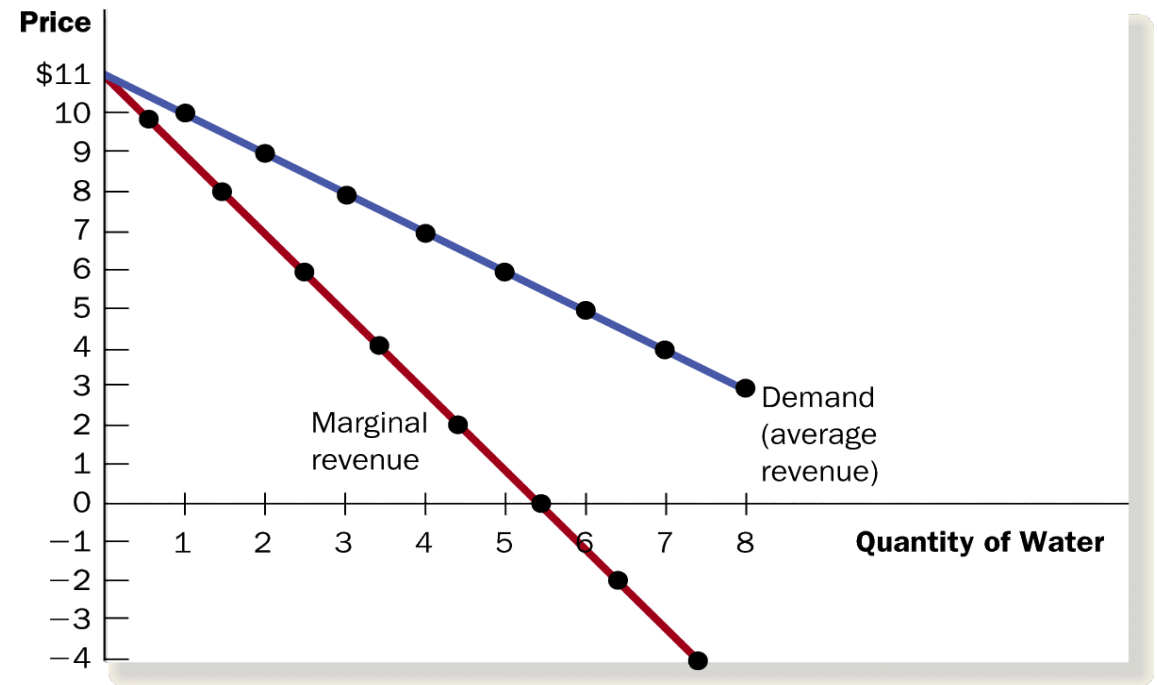
Monopolist's Revenue

- When monopolist raises Q , two effects on TR (same as previously with monopolistic competition):

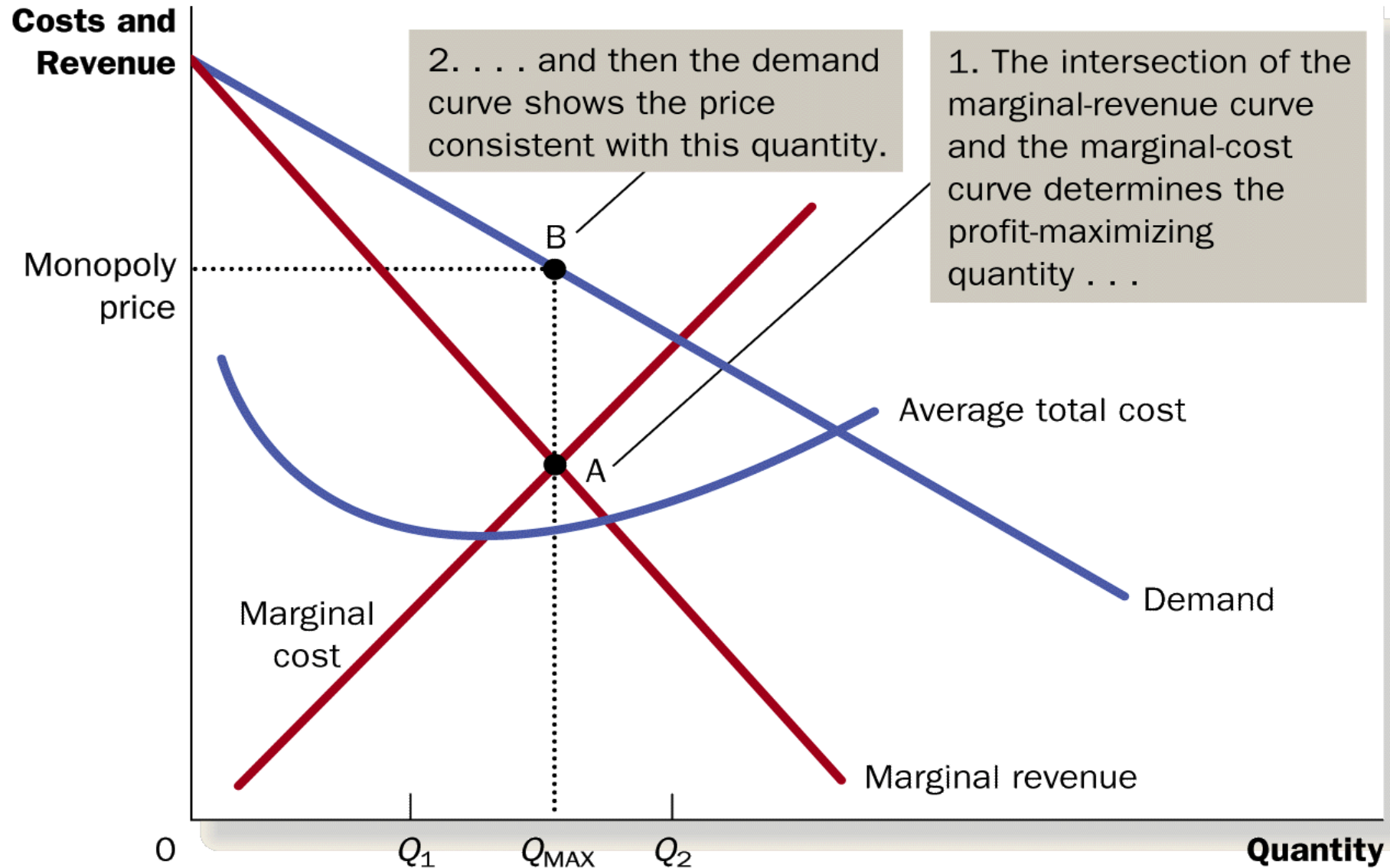
1. Output effect
2. Price effect

Can MR be negative?

Why or why not?



Profit maximization



Profit markup / margin

- Price setting firms maximize profit by choosing the quantity where $MR = MC$
- If demand is downward sloping, then $P > MR$
- Therefore, $P > MC$:
 - There is a price markup / profit margin:

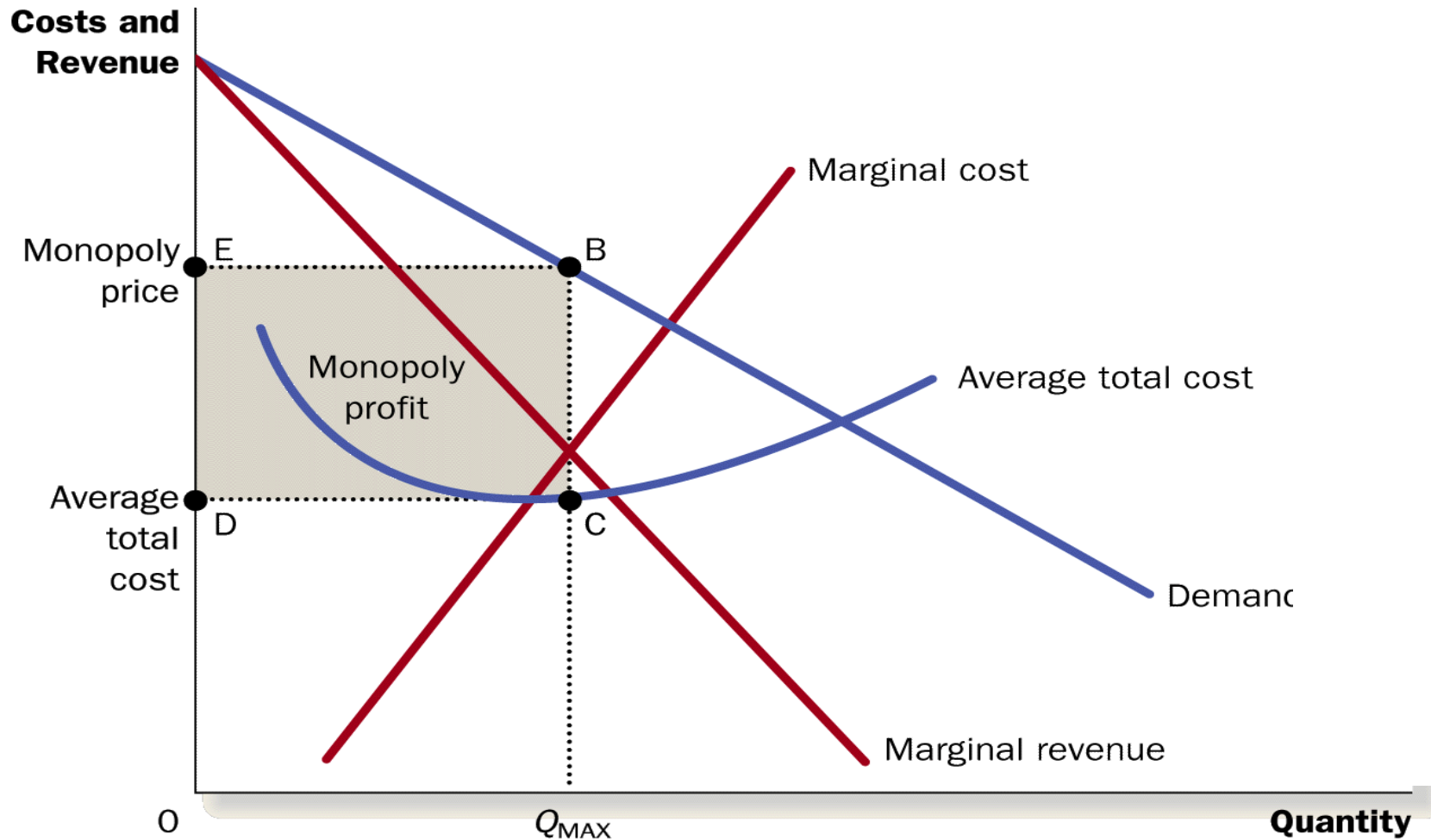
$$P - MC > 0$$

- The profit margin can be expressed as a ratio of the price:

$$\text{profit margin ratio} = \frac{P - MC}{P}$$

Monopoly's profit

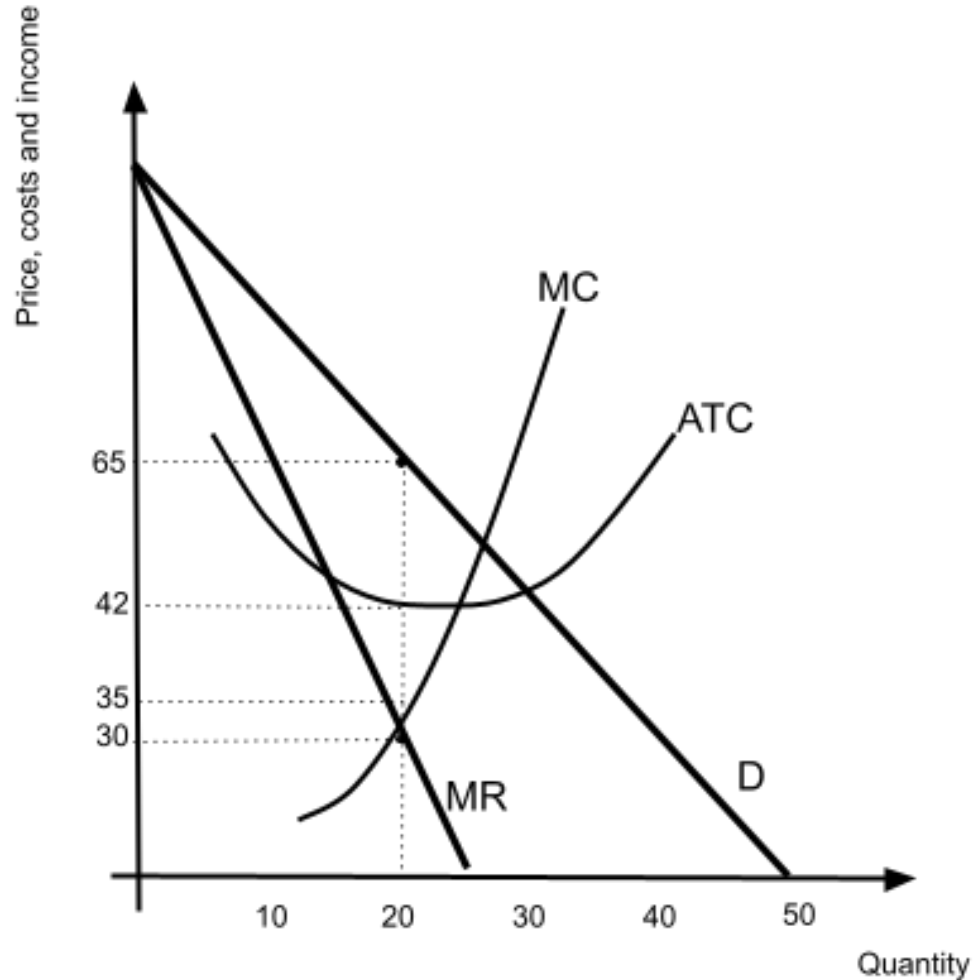
$$\begin{aligned}\text{Profit} &= TR - TC \\ &= P \times Q - TC \\ &= P \times Q - AC \times Q \\ &= Q \times (P - AC)\end{aligned}$$



Is this a good way of organizing a market?

- For the consumers, the higher prices make monopoly undesirable.
- For the owners of the firm, the higher prices make monopoly very desirable.
- Can an objective assessment be made?

Exercise: Monopoly



1. What is the profit maximising quantity?
2. What is the monopoly price?
3. Calculate this monopoly's profit.